



Rational Design of Platinum Anchoring Sites via Stepwise Surface Engineering in MXene

Speaker: Bo-Heng Huang

Advisor: I-Wen Peter Chen

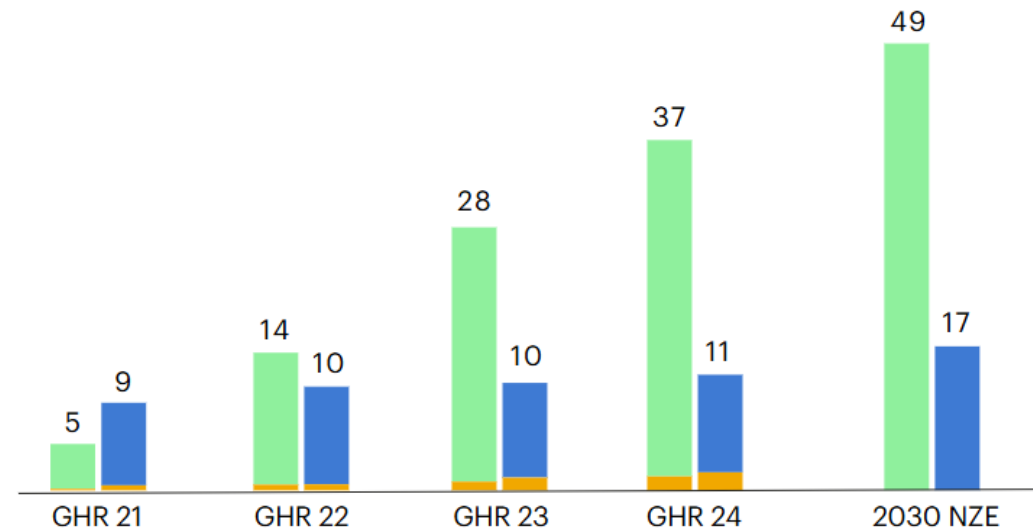
Date: April 16, 2026



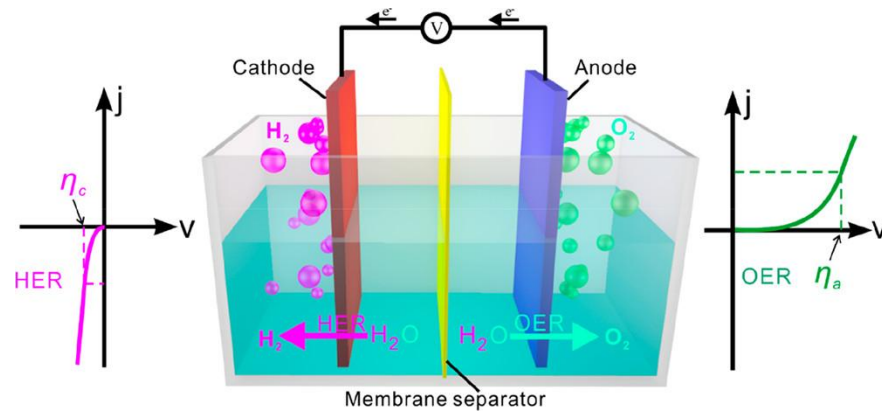
Low-emissions hydrogen production from announced projects by 2030

Mtpa

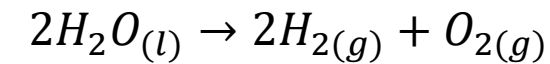
● Renewables ● Fossil fuels with CCUS ● FID



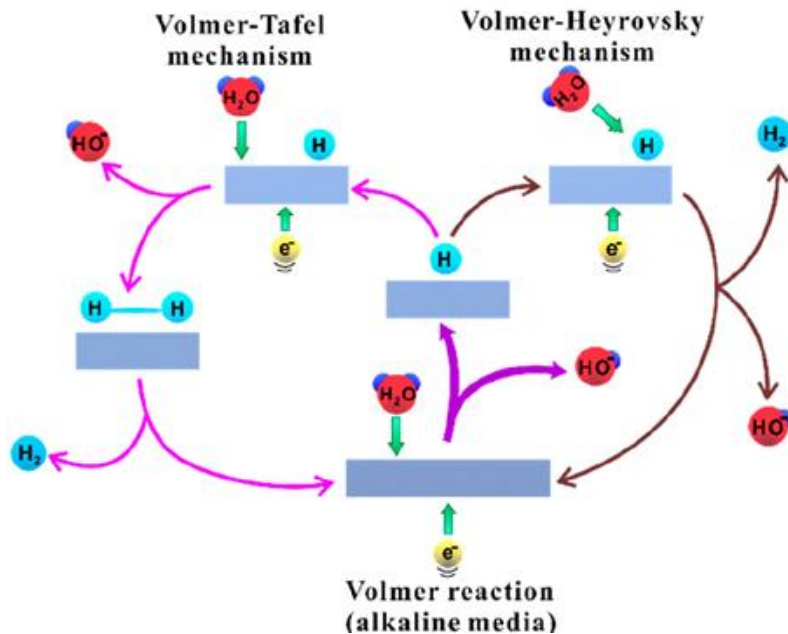
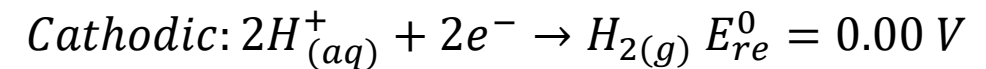
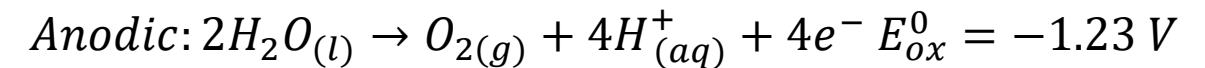
Global hydrogen demand reached more than 97 Mt in 2023 and could reach almost 100 Mt in 2024.

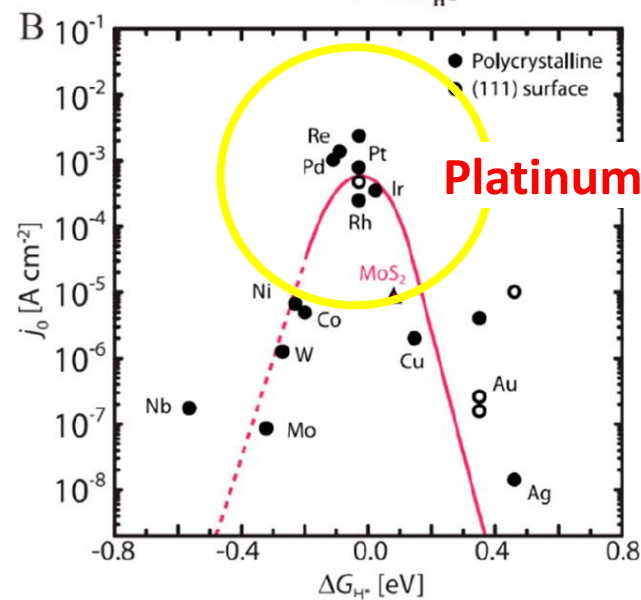
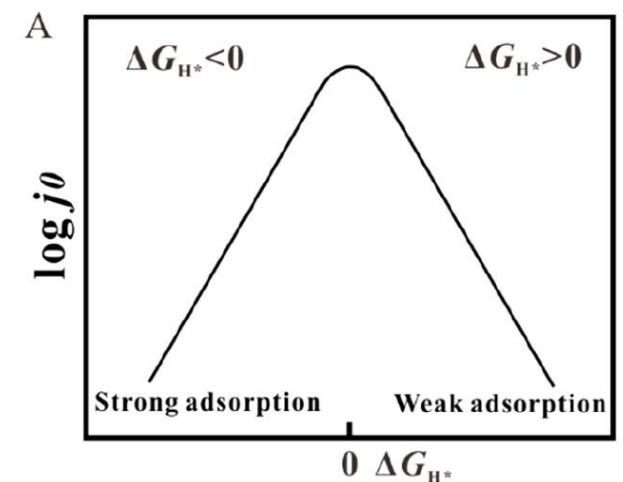


Overall reaction:



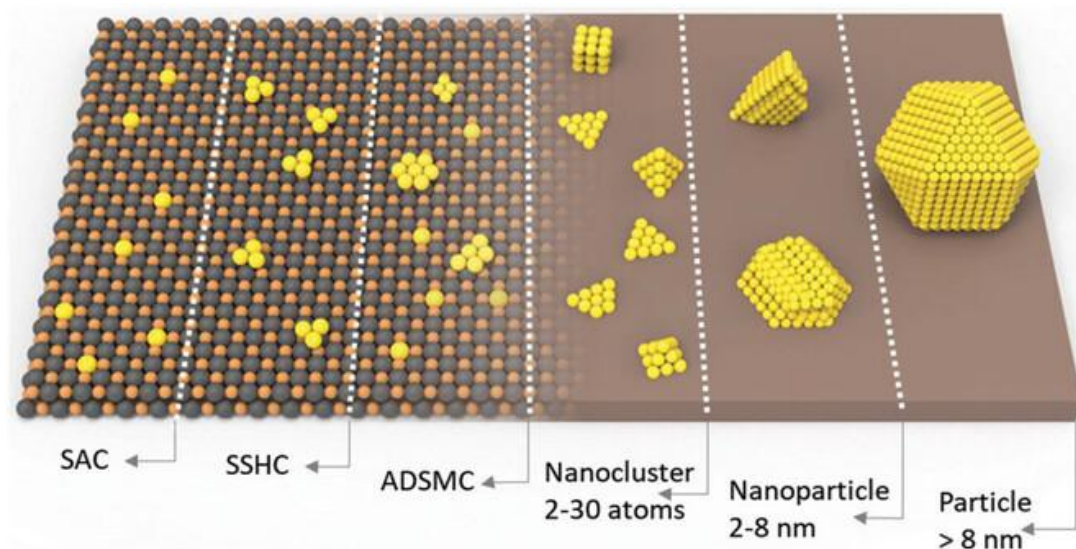
Reaction in acidic solution:



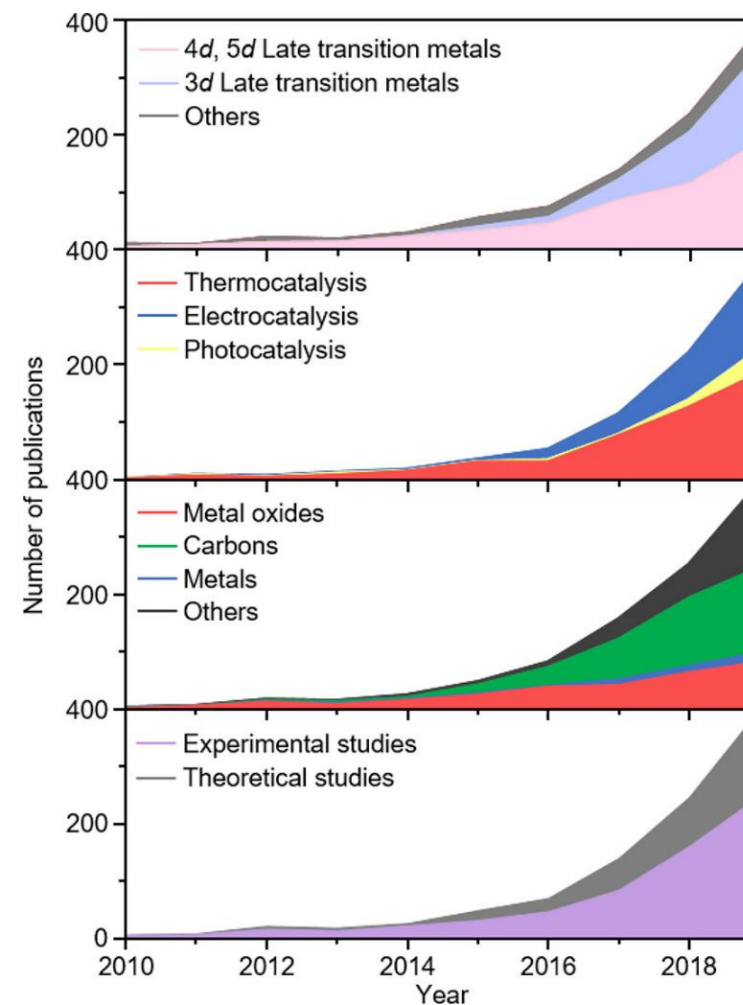


Challenge:

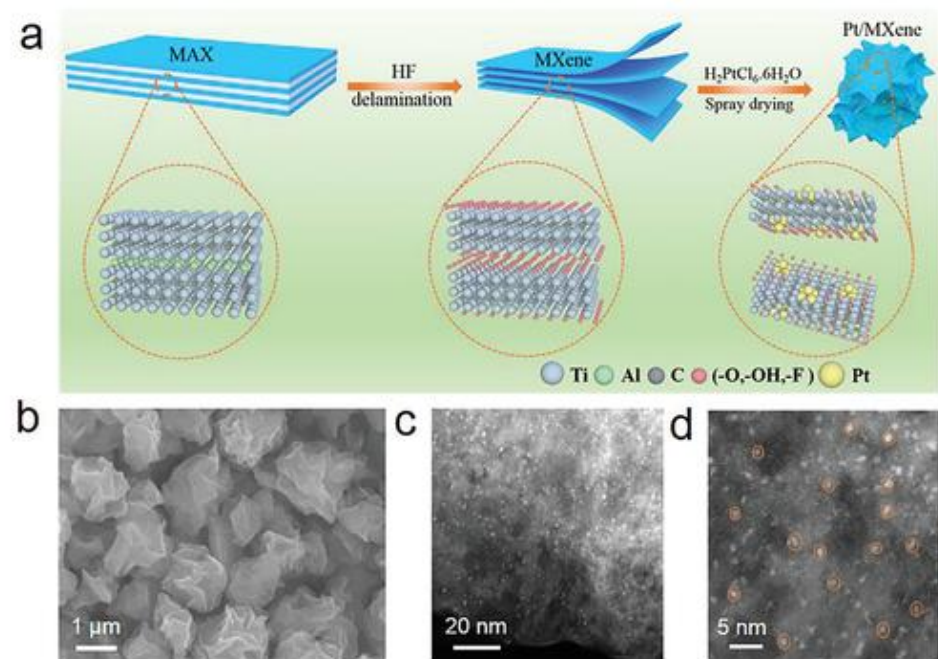
1. High costs
2. Low abundance
3. Insufficient stability



1. Metal oxides
2. Metal (Single-Atom Alloys, SAAs)
3. Carbon-based Materials

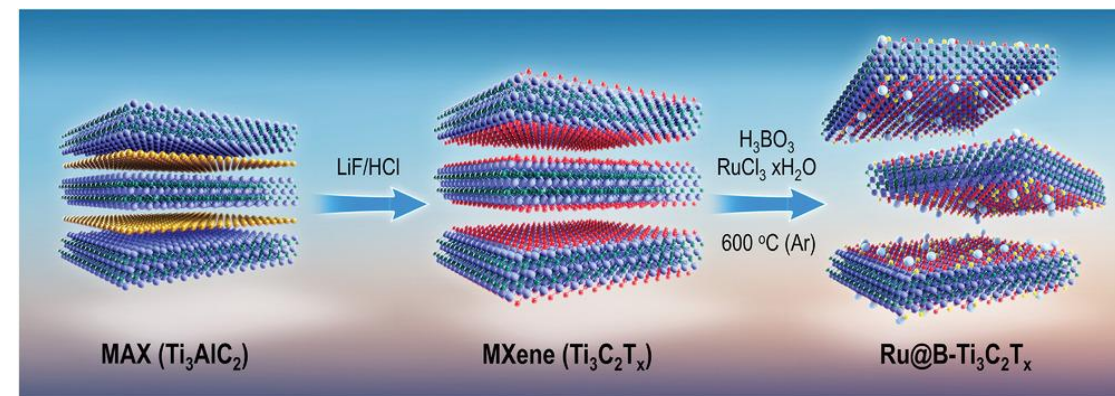


Heterogeneous supports : MXene

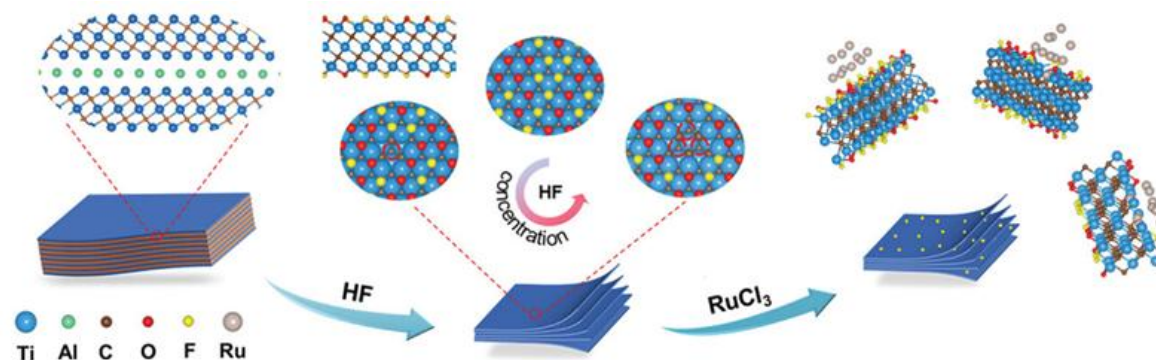


1. Electronic conductivity
2. Rich surface terminal groups
3. Good hydrophilicity
4. Open layered structure

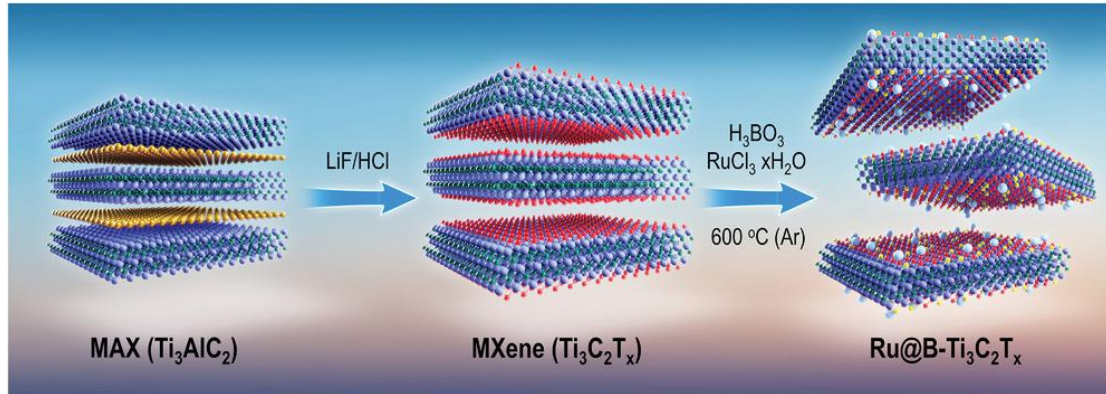
1. Termination engineering



2. Vacancy engineering



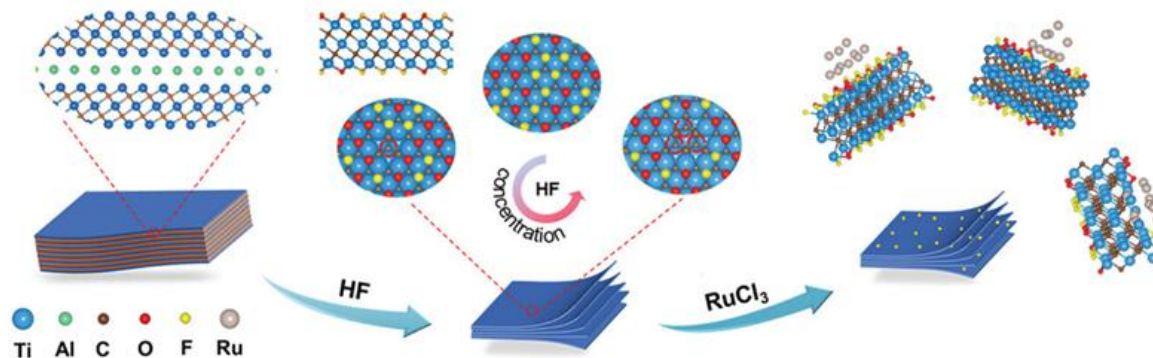
1. Termination engineering



Calcination

High Energy Consumption

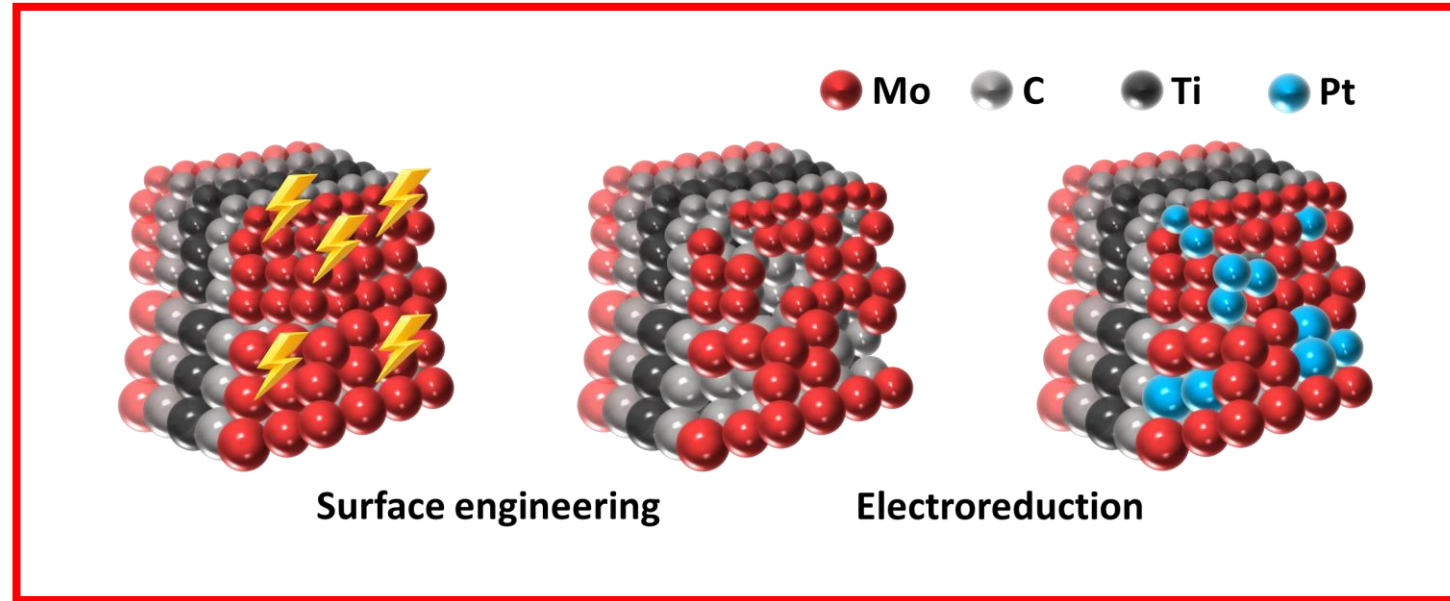
2. Vacancy engineering



High-Concentration HF

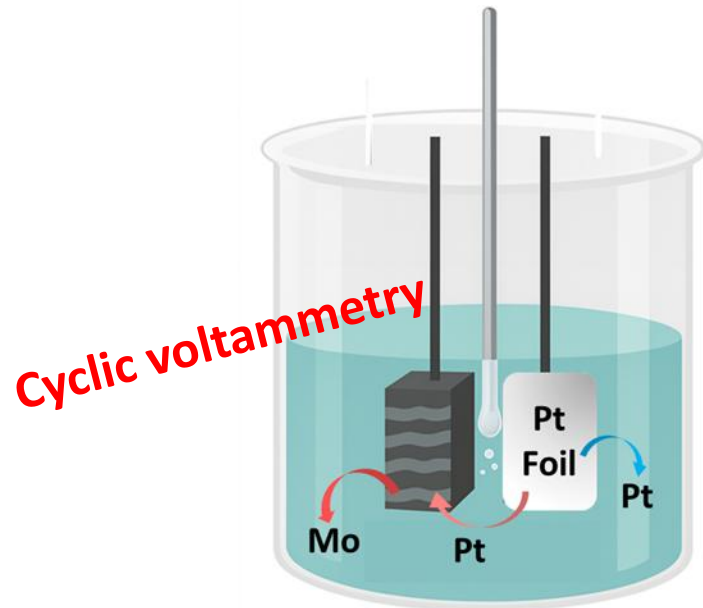
Severe Safety Hazards

Electrochemical deposition: Safer / Precise control



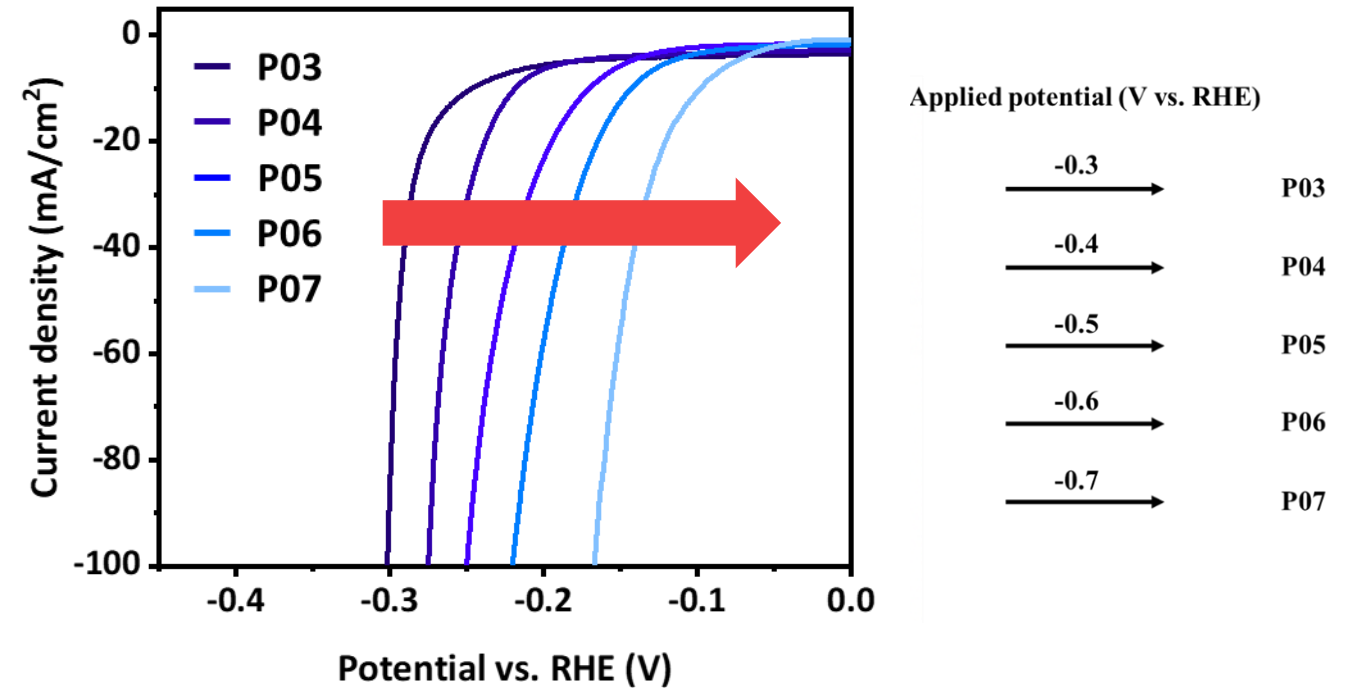
1. How the applied potential modulates the electrochemical deposition?
2. Which specific surface configurations most favor the electrochemical deposition and anchoring of Pt single atoms?

Setting:

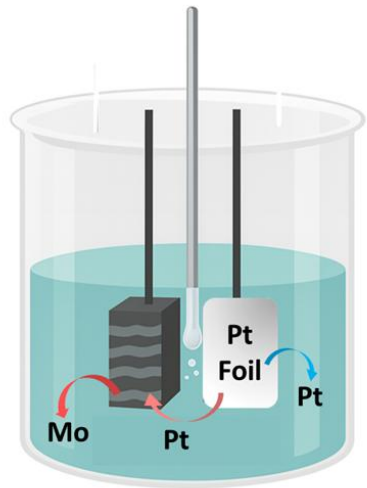


Tuning potential to obtain different performances of $\text{Mo}_2\text{TiC}_2\text{T}_x$

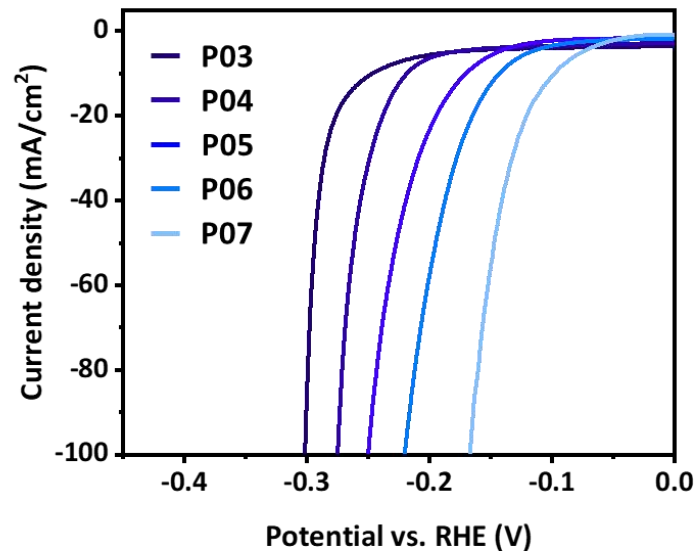
LSV curves:
HER performance



A more negative reduction potential applied during preparation leads to enhanced HER catalytic activity.

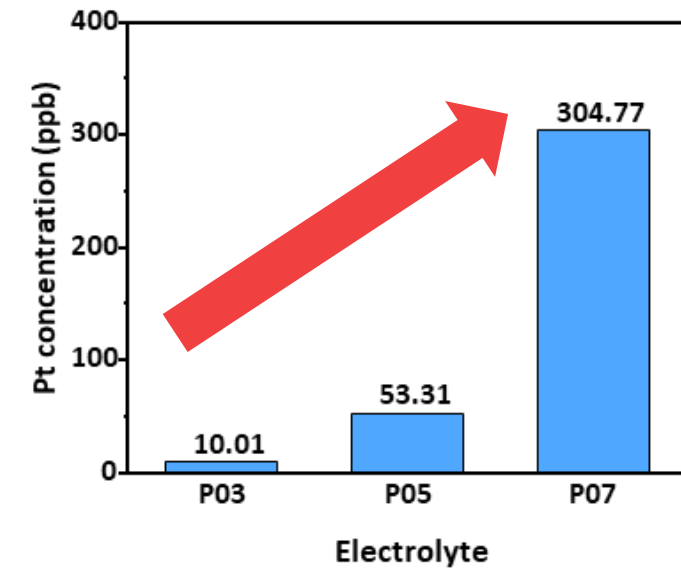


Tuning potential to obtain different performances of $\text{Mo}_2\text{TiC}_2\text{T}_x$



ICP-MS:

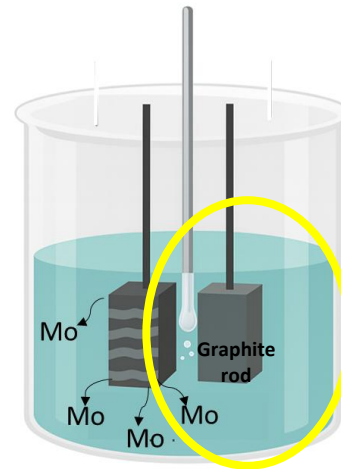
Pt concentration in electrolyte



The leaching of Pt from the counter electrode becomes more significant when the system is subjected to a highly negative potential.

Step 1 Carving

Surface engineering

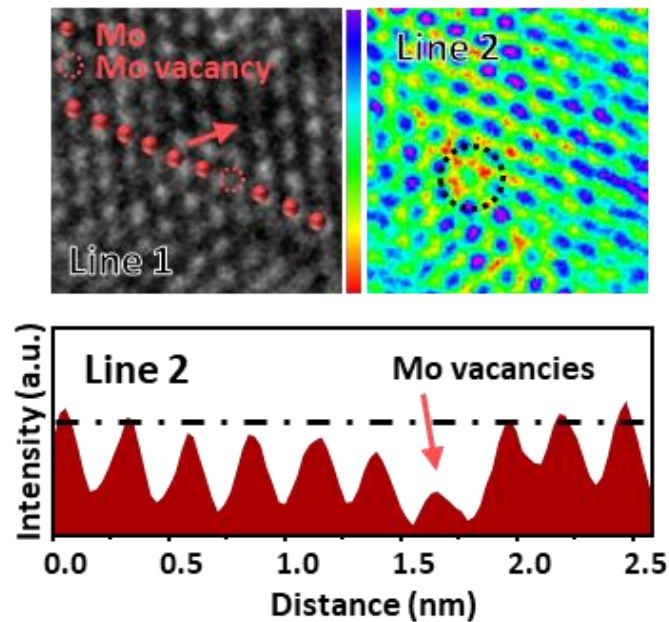


Potential-tuned vacancies
and functional groups

More vacancy

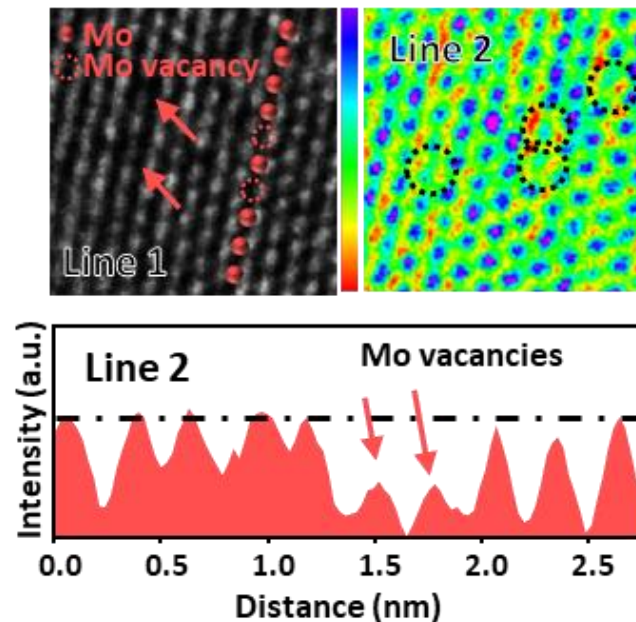
HR-TEM:

Low potential treated (C03)



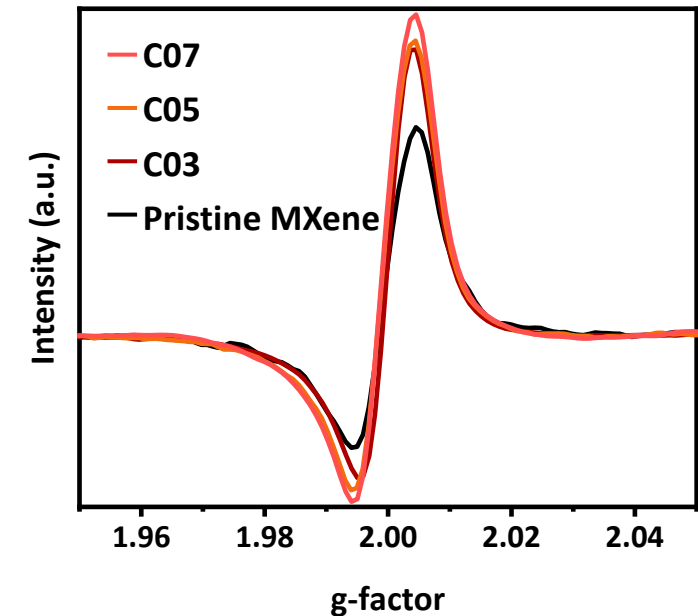
HR-TEM :

High potential treated (C07)



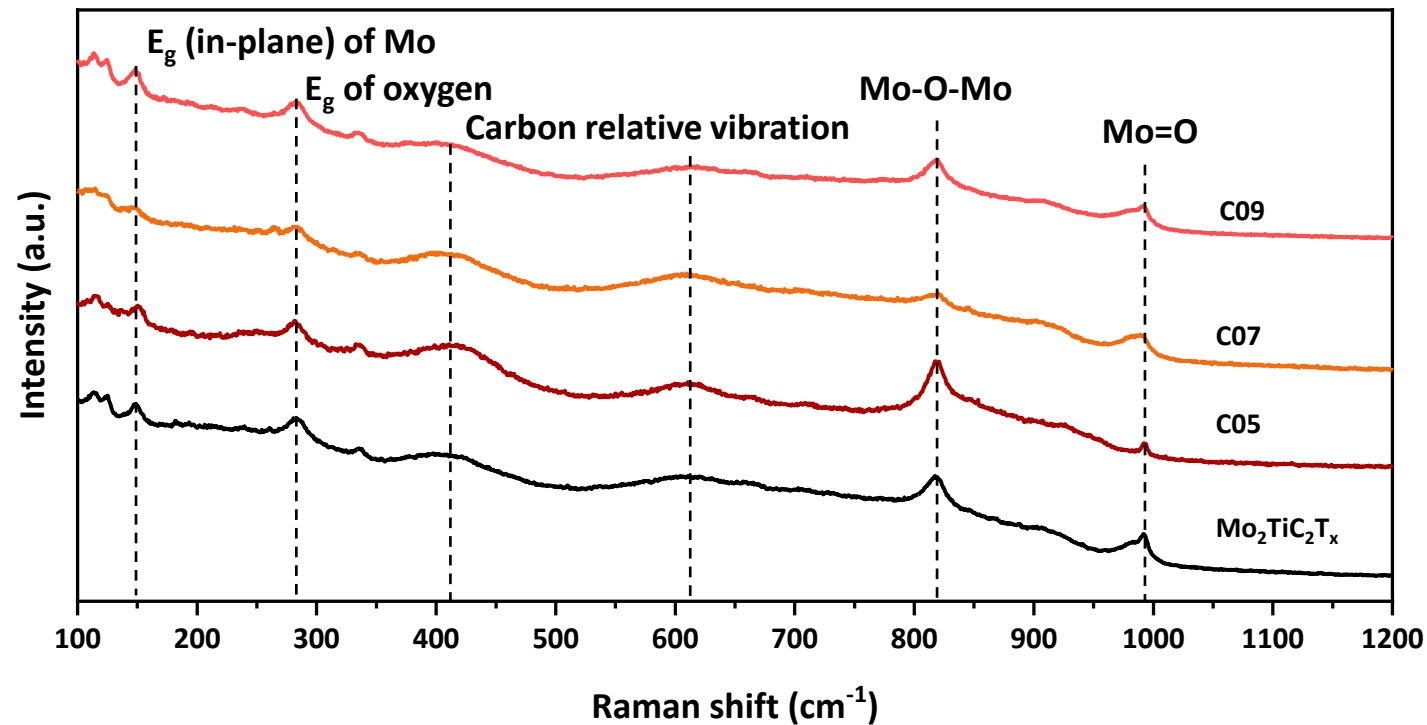
EPR:

Defect signal



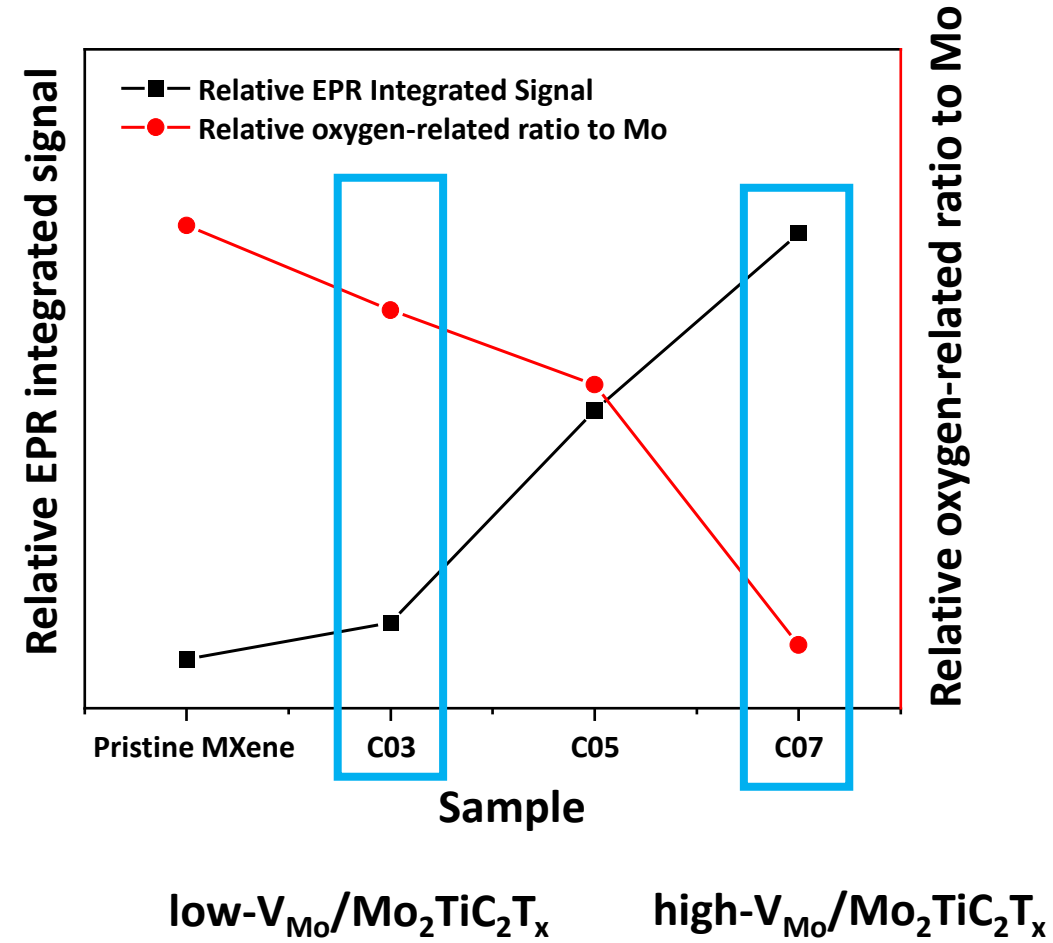
Applying a more negative potential leads to the formation of more vacancies on the MXene surface.

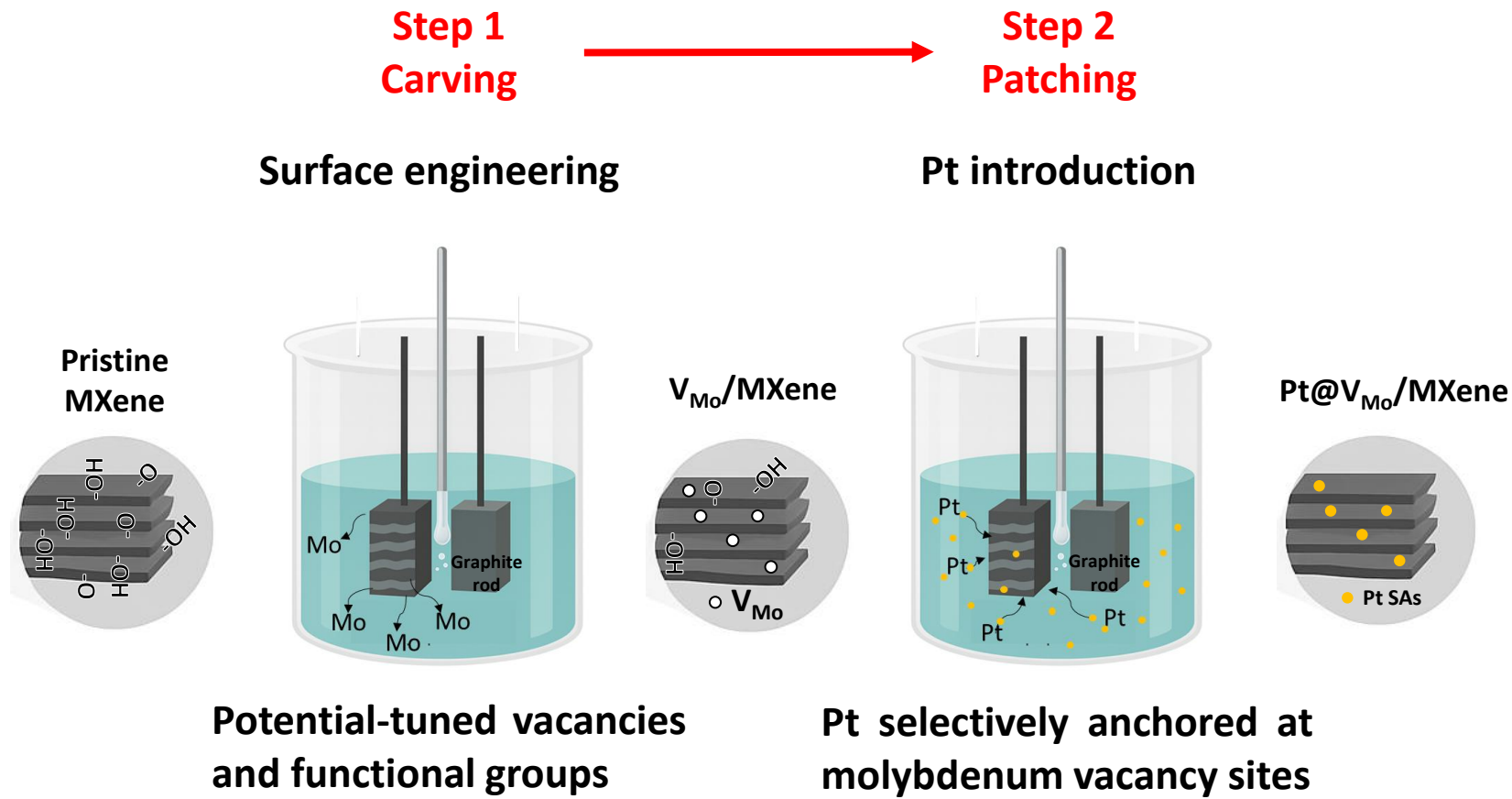
Raman:
Functional groups change



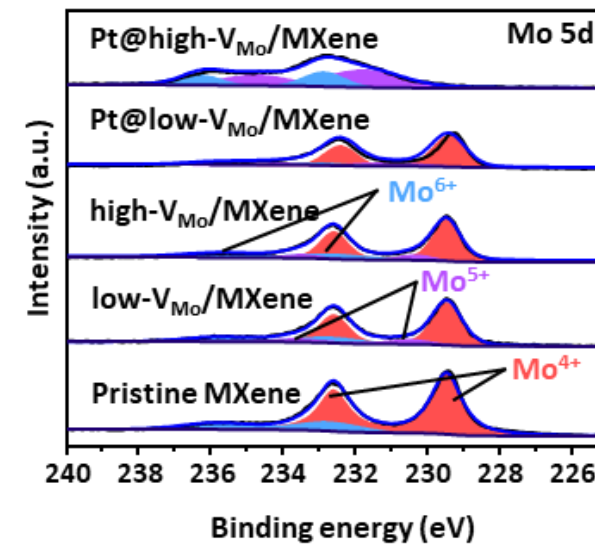
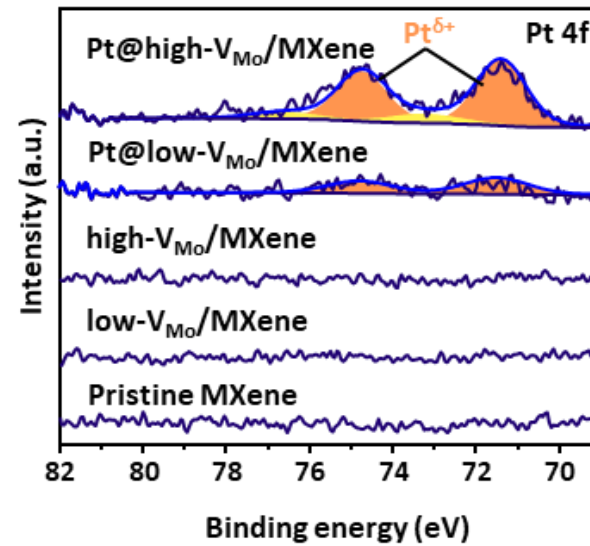
Mode	Raman shift	Pristine	C03	C05	C07
E_g in-plane vibration of Mo atom	148 cm^{-1}	89188.6	42809.8	61153.8	78686.7
E_g vibration of the O atoms	277 cm^{-1}	161118	73645.7	103863	124551
Oxygen functionalization	818 cm^{-1}	98238.2	48179.4	62122.9	73070.8
	985 cm^{-1}	37293.6	16992.4	27825.2	31540
Relative oxygen-related ratio to Mo		3.3261	3.2426	3.1692	2.9123

Lowest !!!





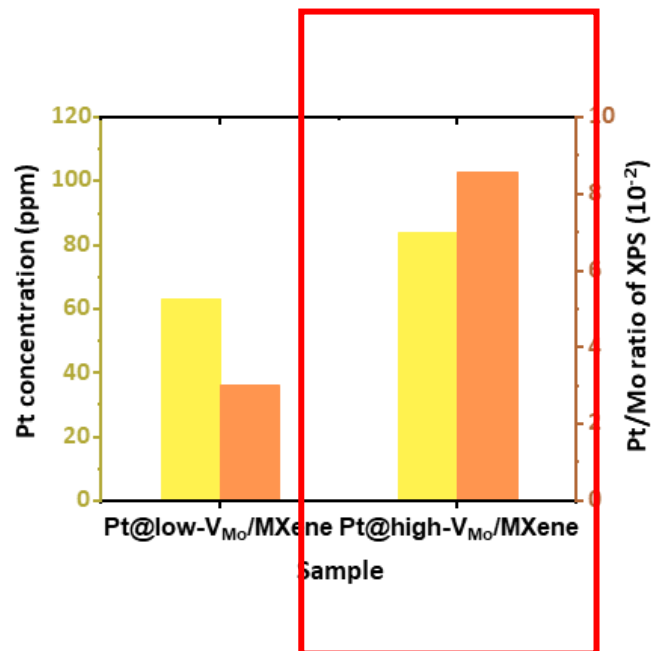
XPS:
Charge redistribution



The absence of metallic Pt⁰ signals or satellite peaks associated with clustered Pt corroborates the single-atom dispersion.

ICP-MS:

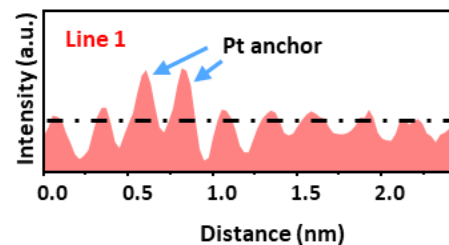
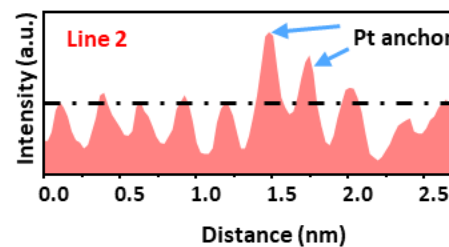
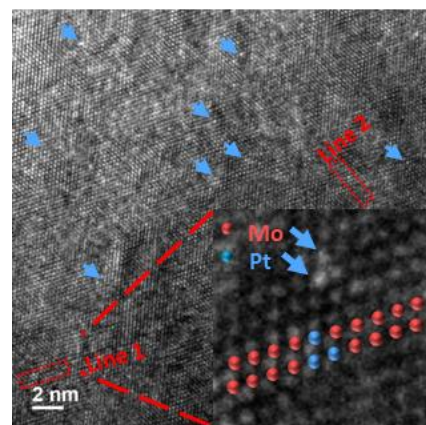
Pt concentration on working electrode



Vacancies domain !!!

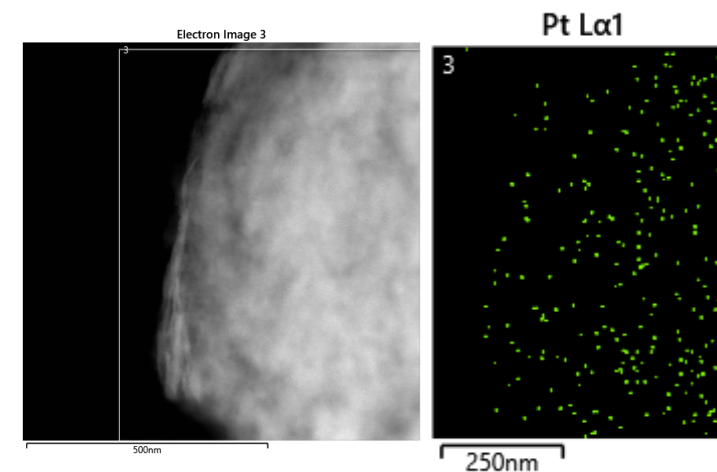
HR-TEM:

Pt anchor on vacancies sites

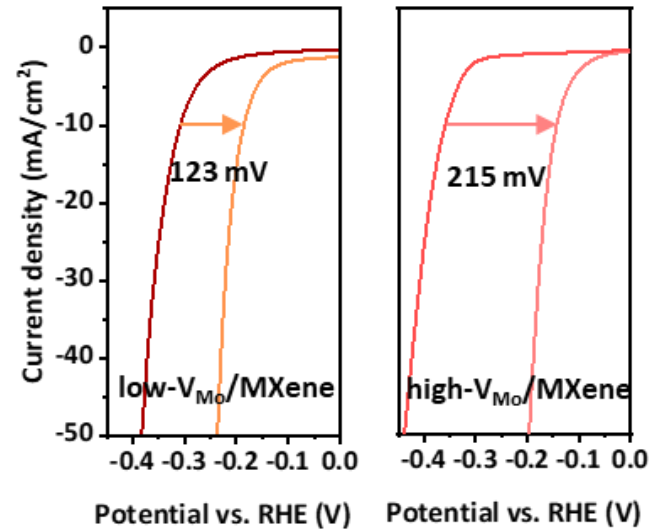


EDS:

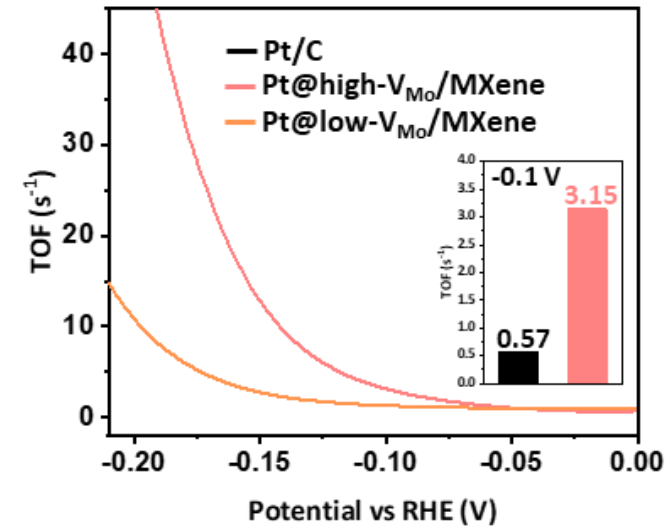
Great dispersion



LSV curves:

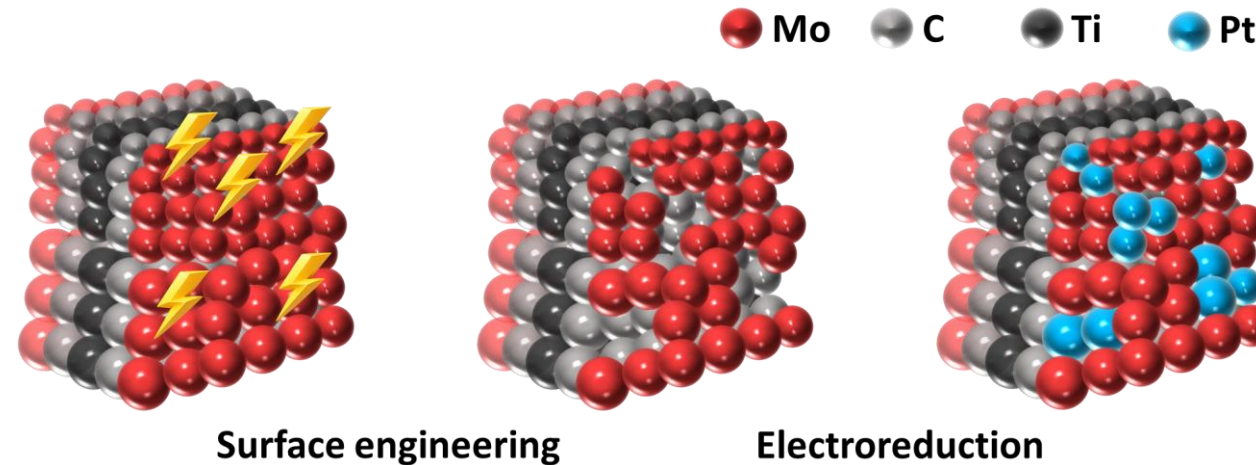


TOF:



Conclusion

1. By decoupling surface engineering from metal embedding, this strategy achieves precise control over the anchoring environment, allowing for high-density, site-specific Pt embedding within the $\text{Mo}_2\text{TiC}_2\text{T}_x$ framework.
2. This research elucidates potential-induced interface charge redistribution and defect-mediated migration.



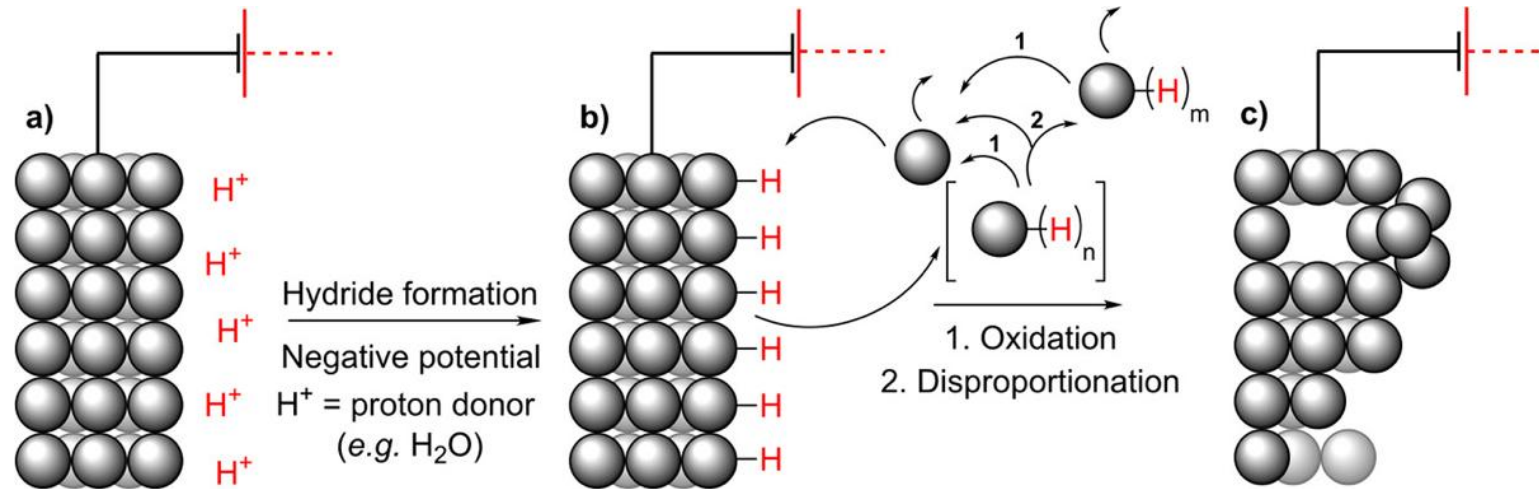
Acknowledge



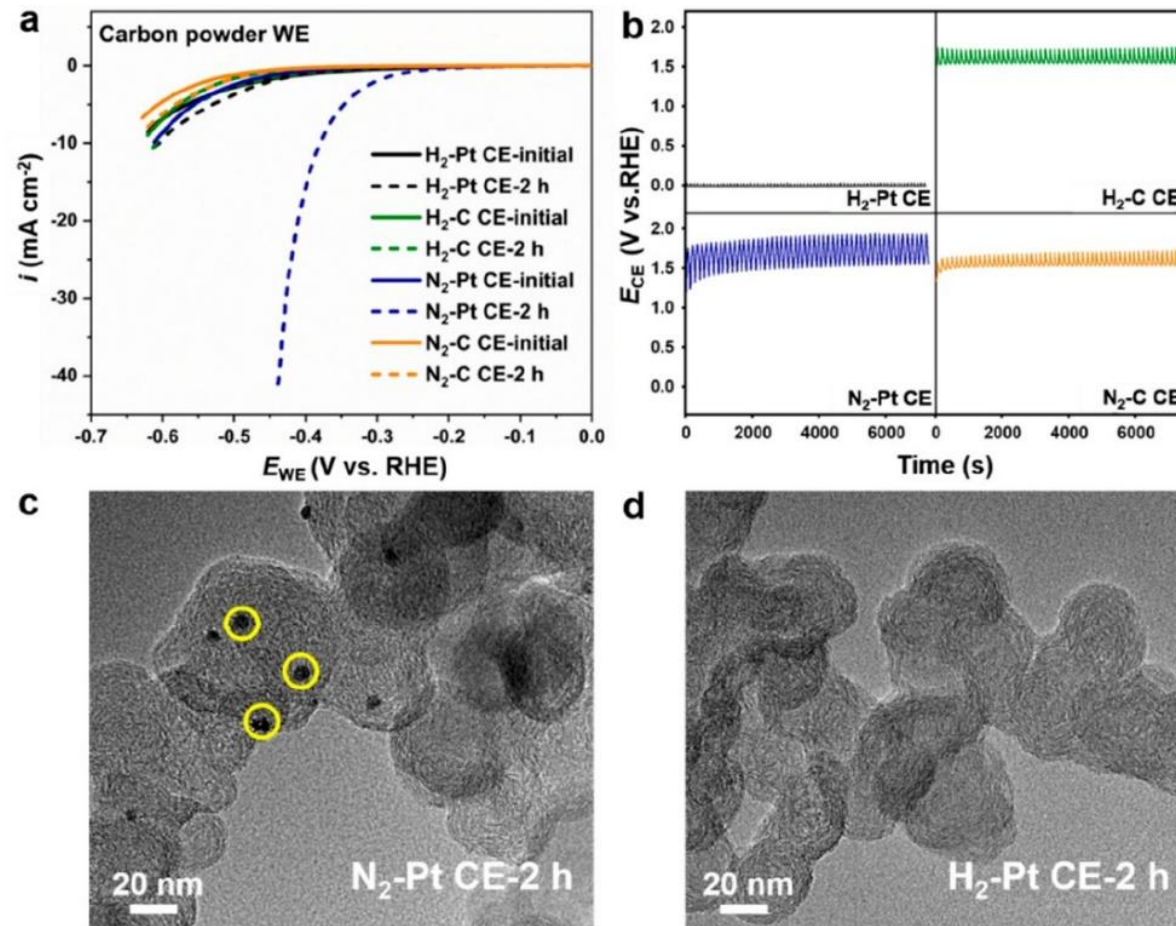


Thanks for your attention

Speaker: Bo-Heng Huang

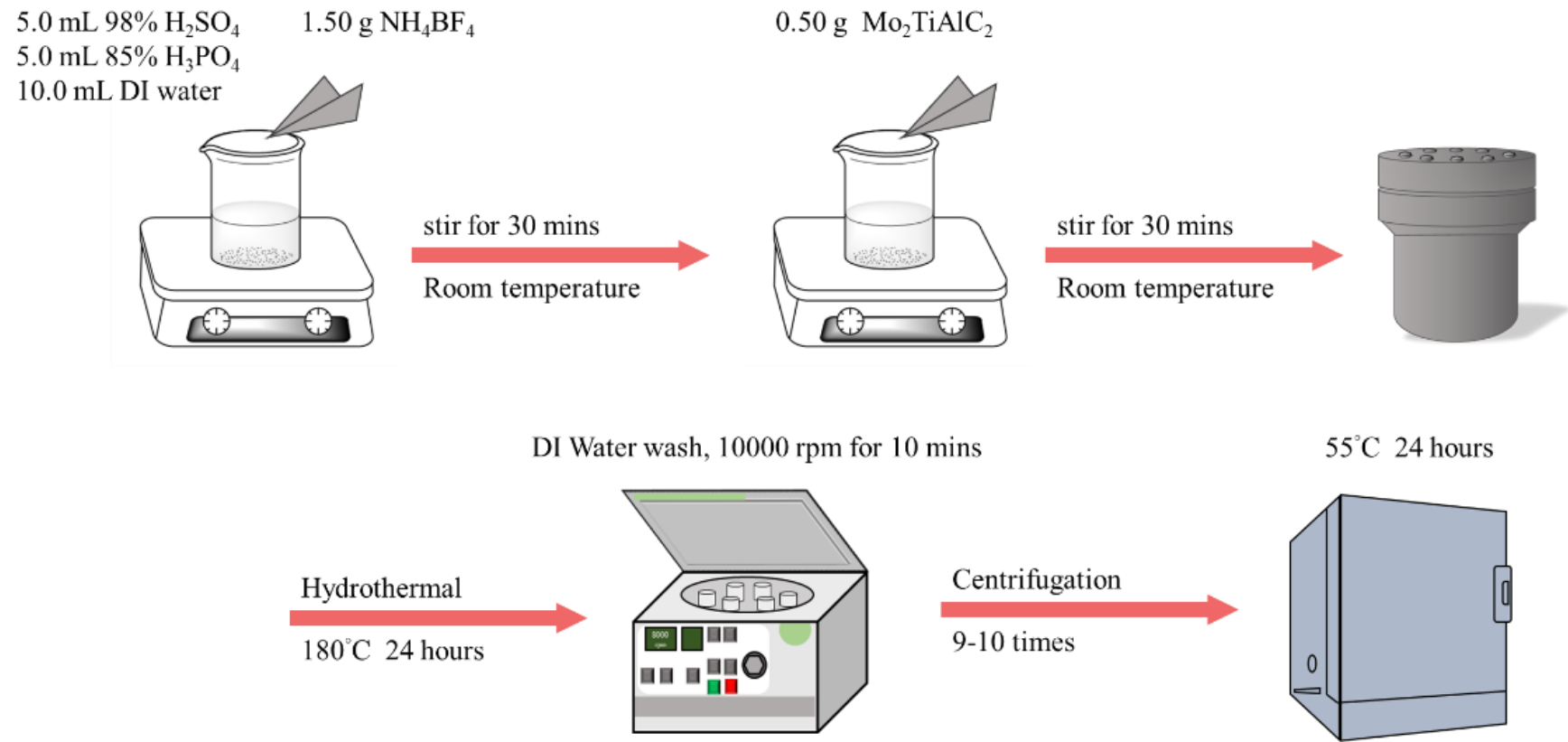


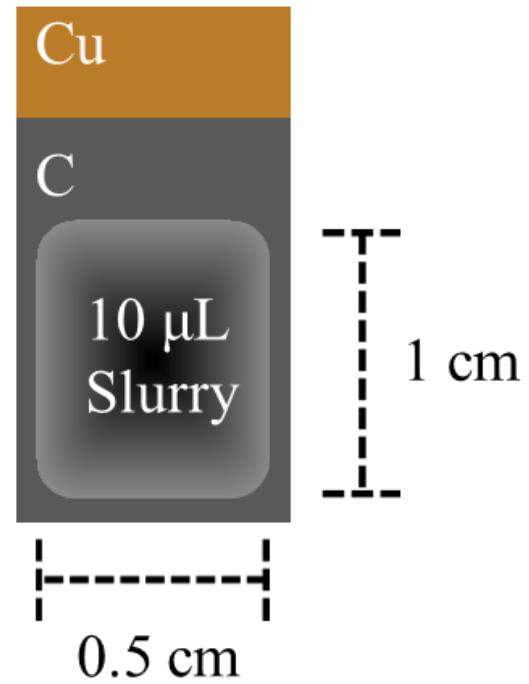
Chem. Rev., 121 (17), 10241–10270, **2021**.

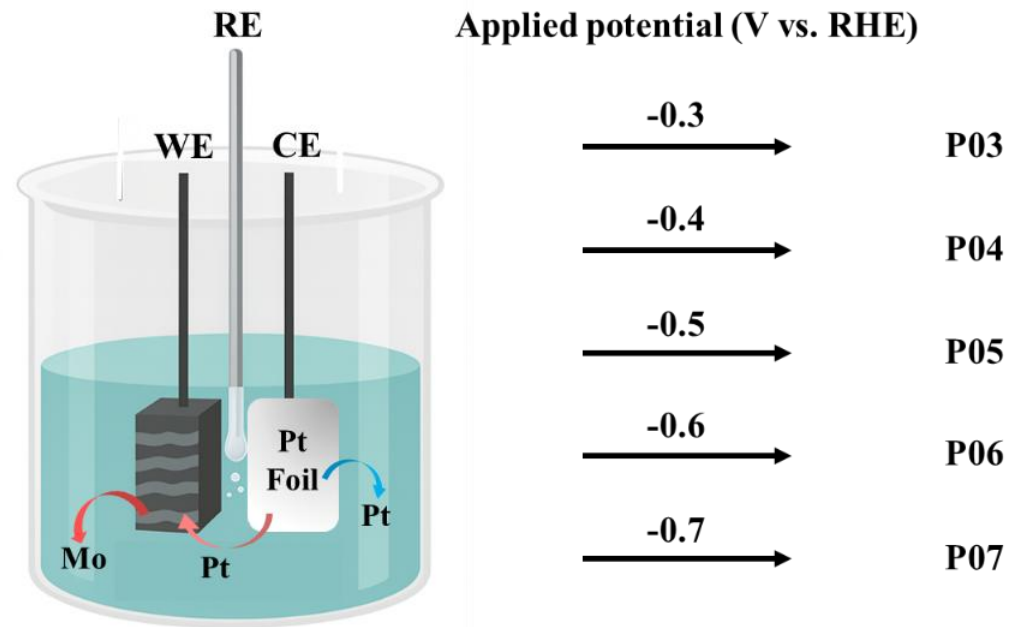


ACS Catal., 13 (4), 2534–2541, **2023**.

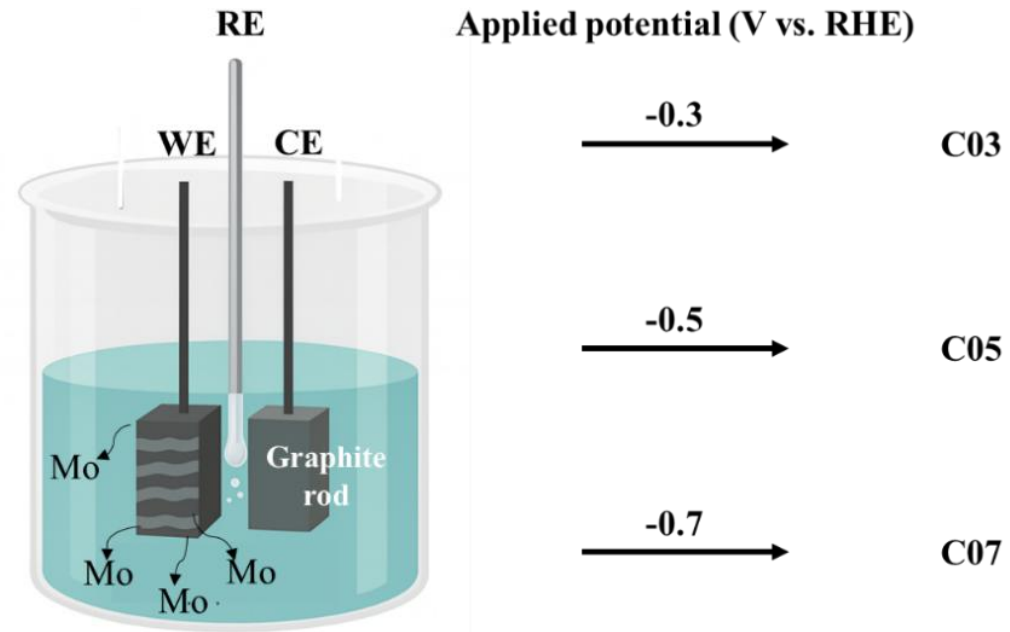
Supporting

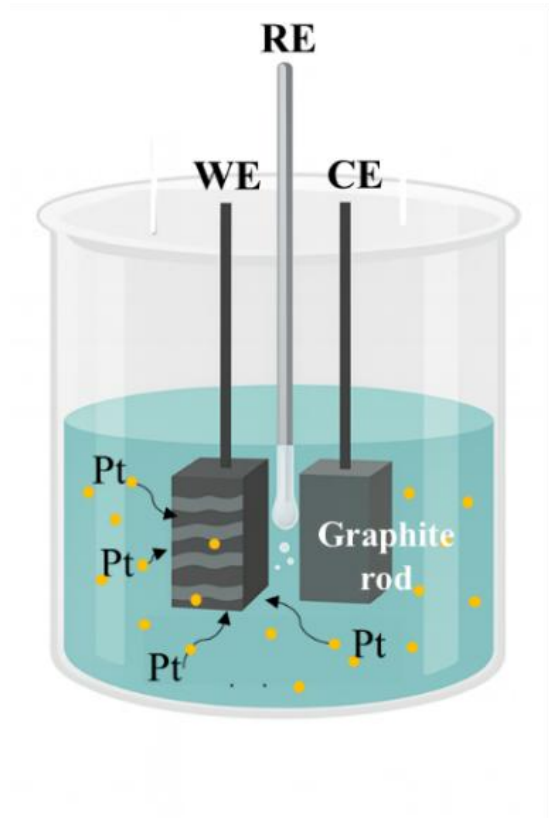






Supporting





Applied potential (V vs. RHE)

C03

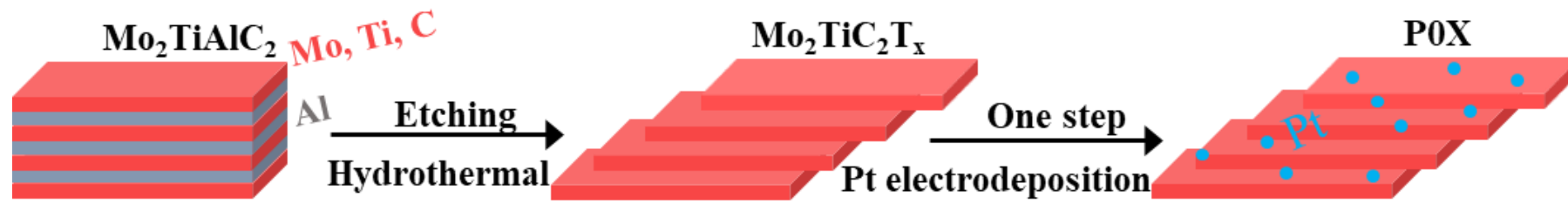
-0.3

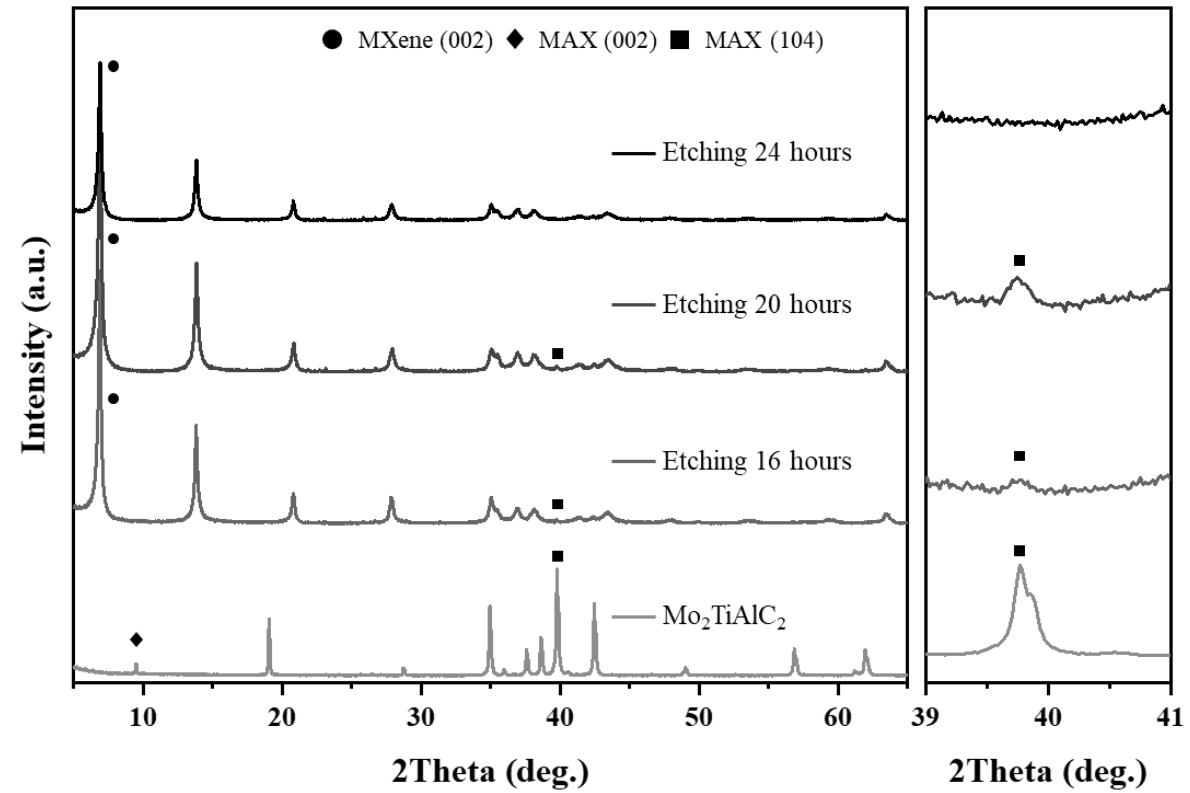
Pt@low- $V_{M0}/Mo_2TiC_2T_x$

C07

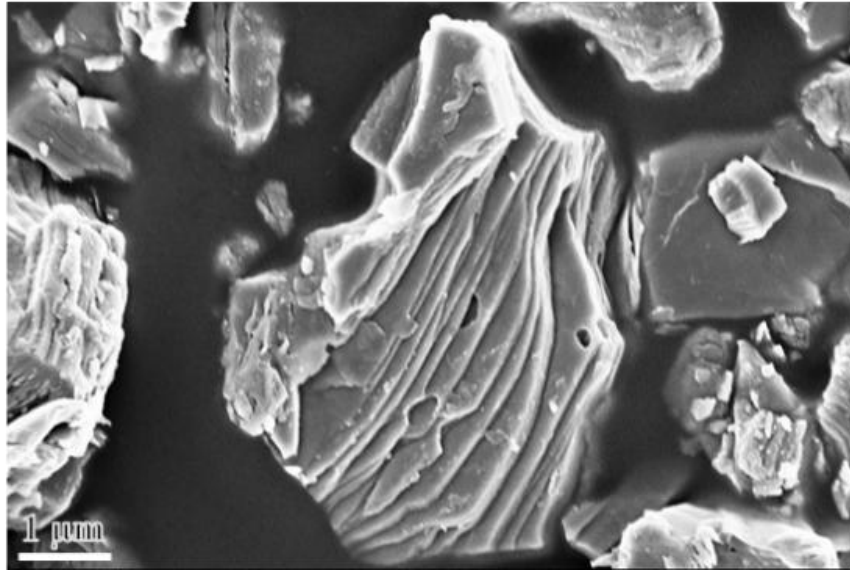
-0.3

Pt@high- $V_{M0}/Mo_2TiC_2T_x$

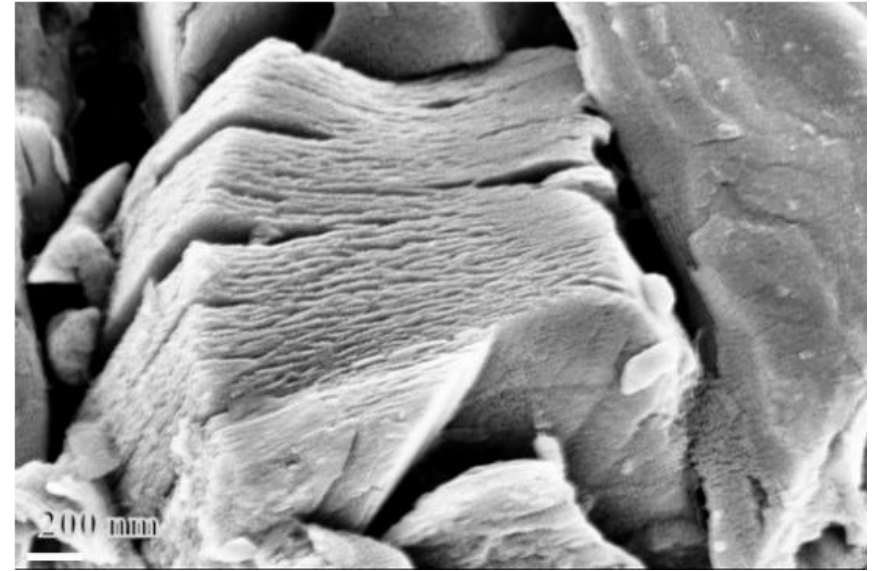


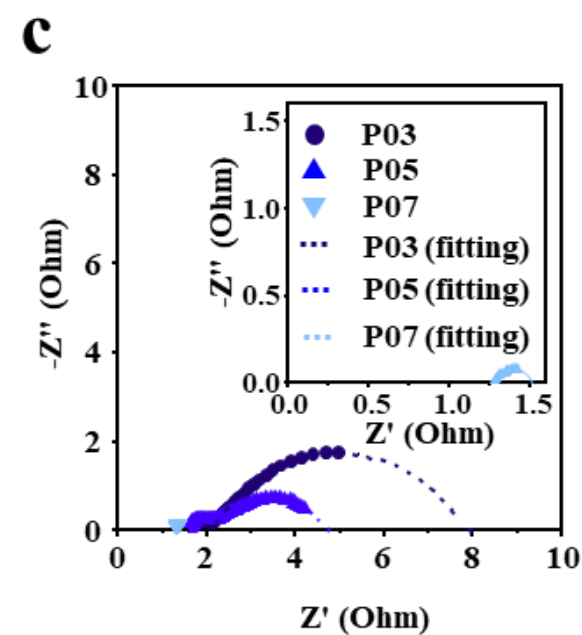
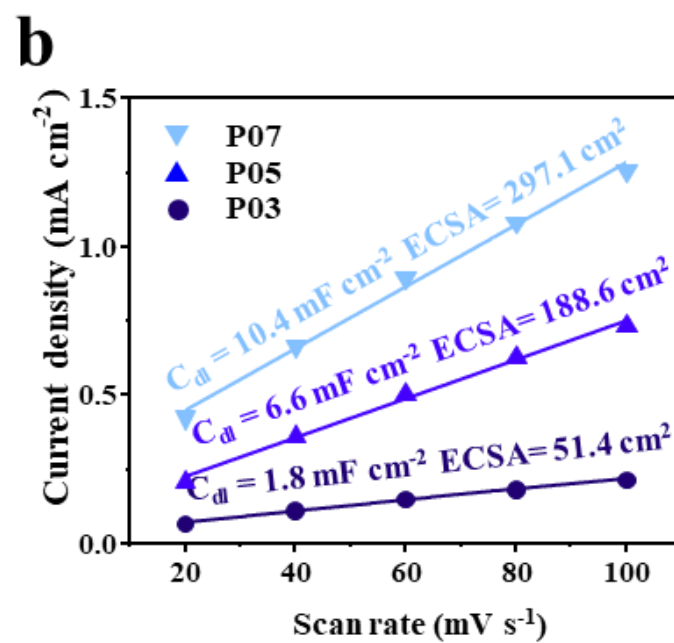
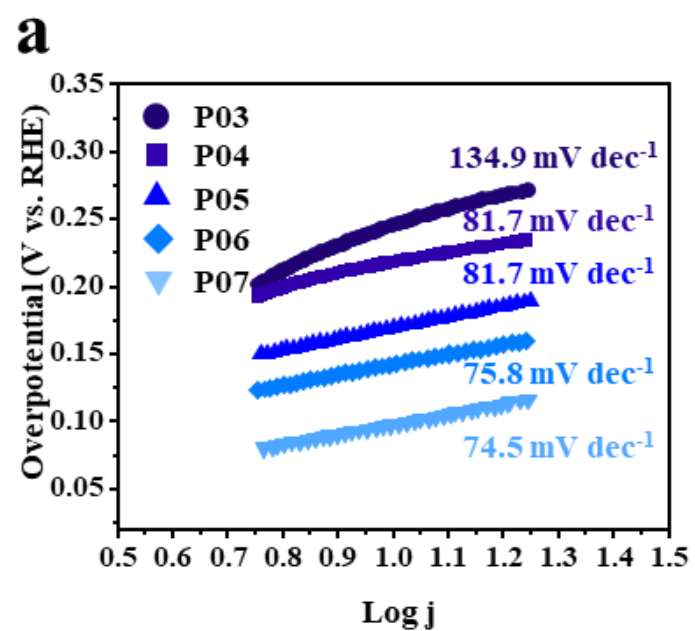


a

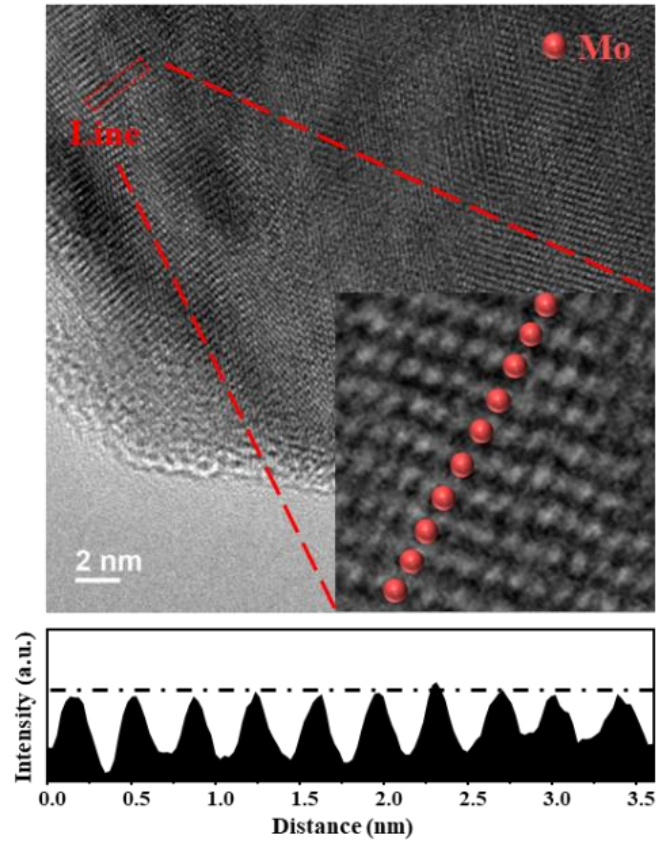


b

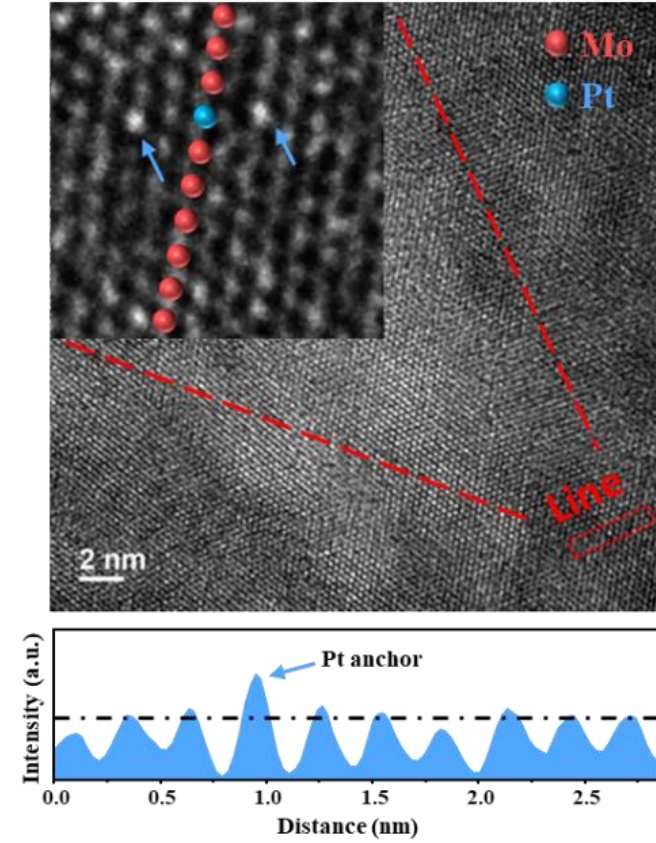


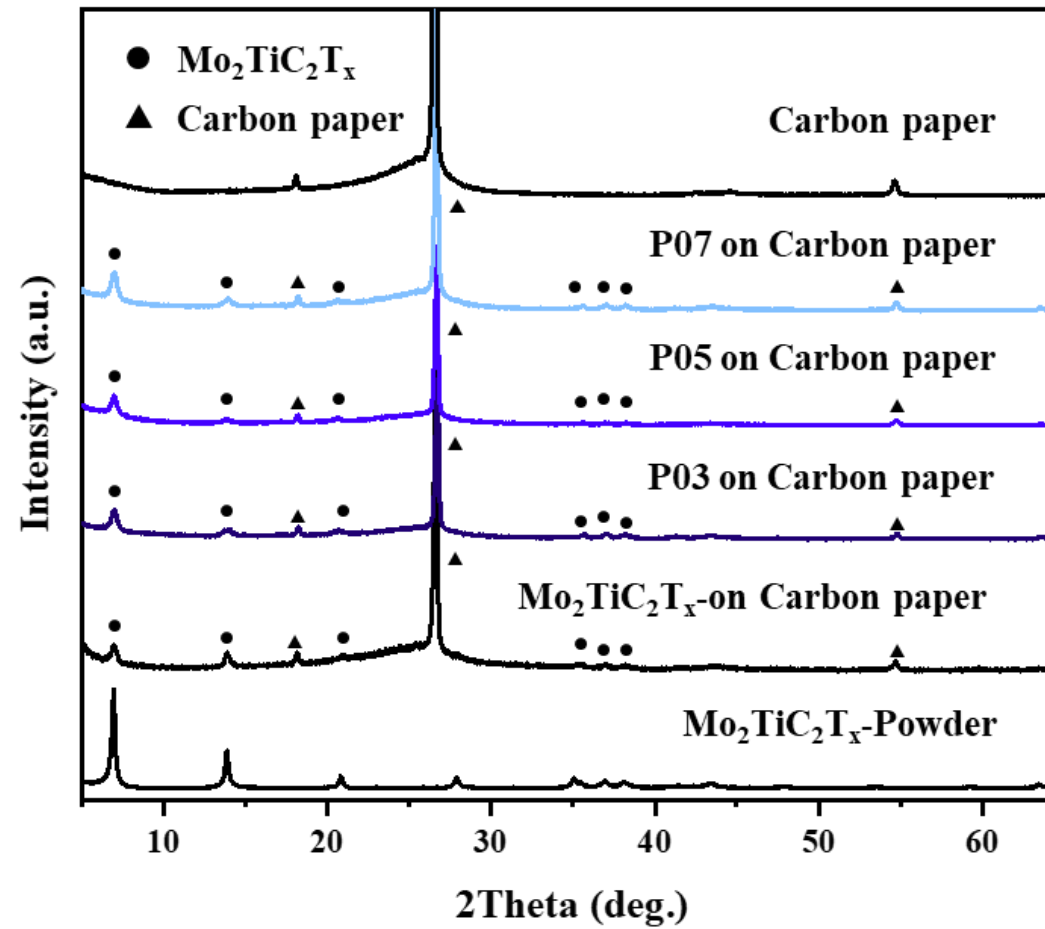


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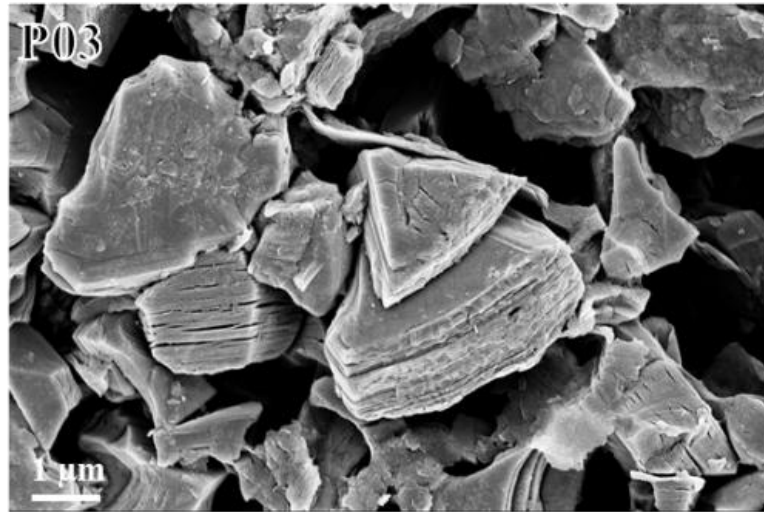


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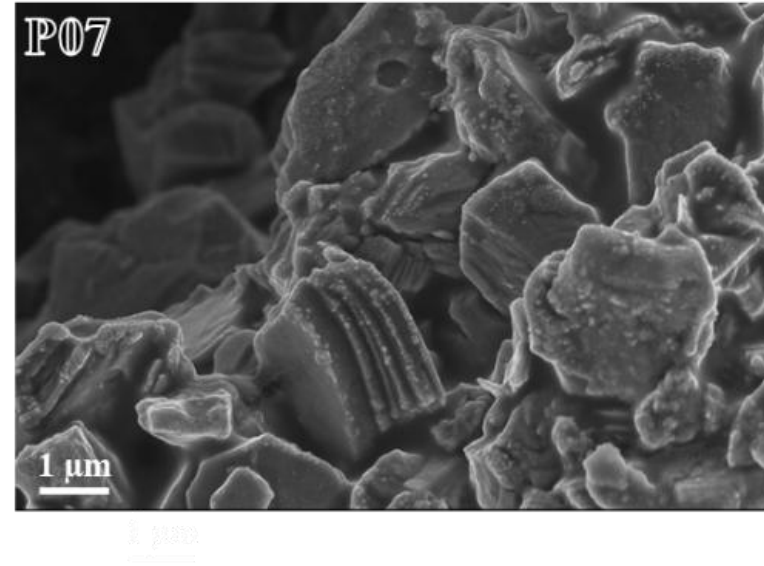


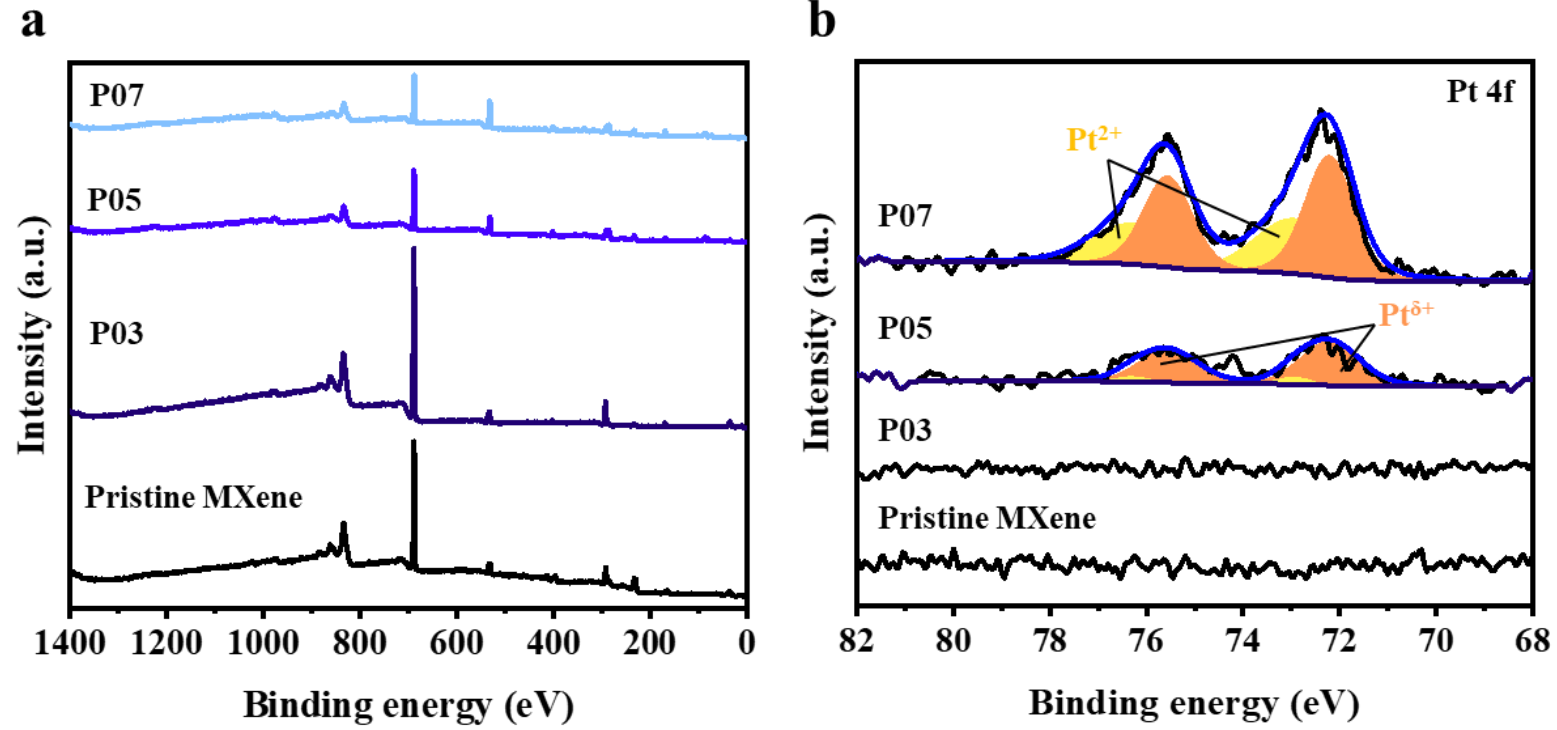


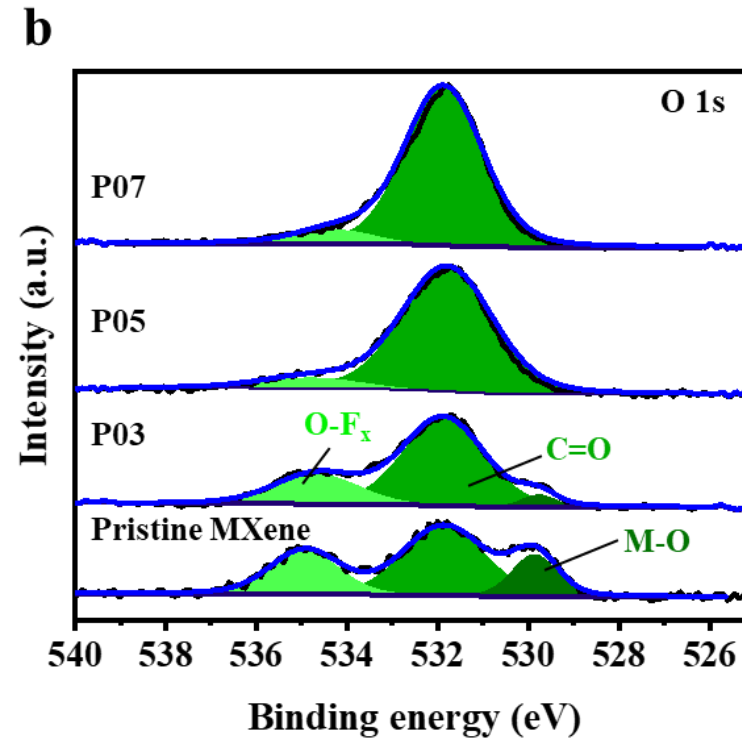
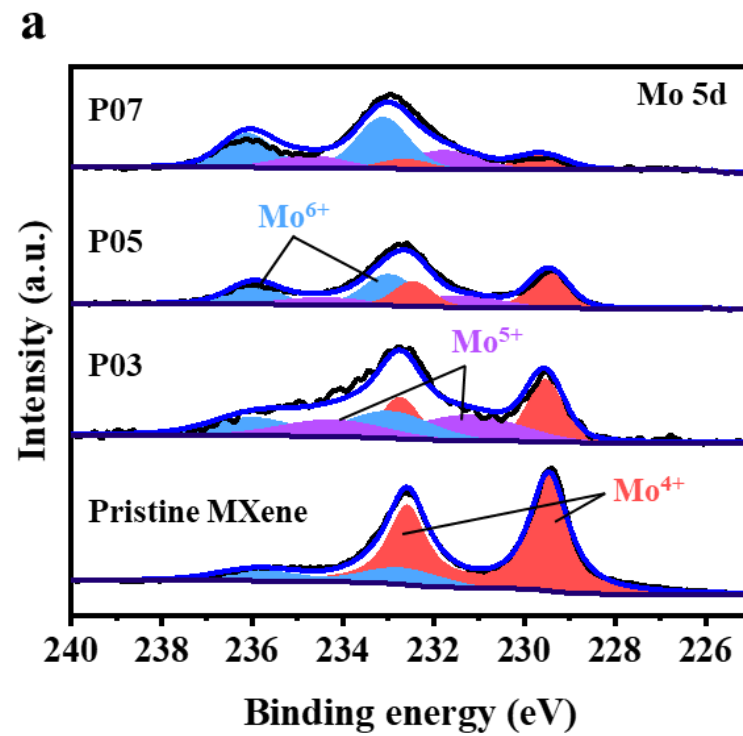
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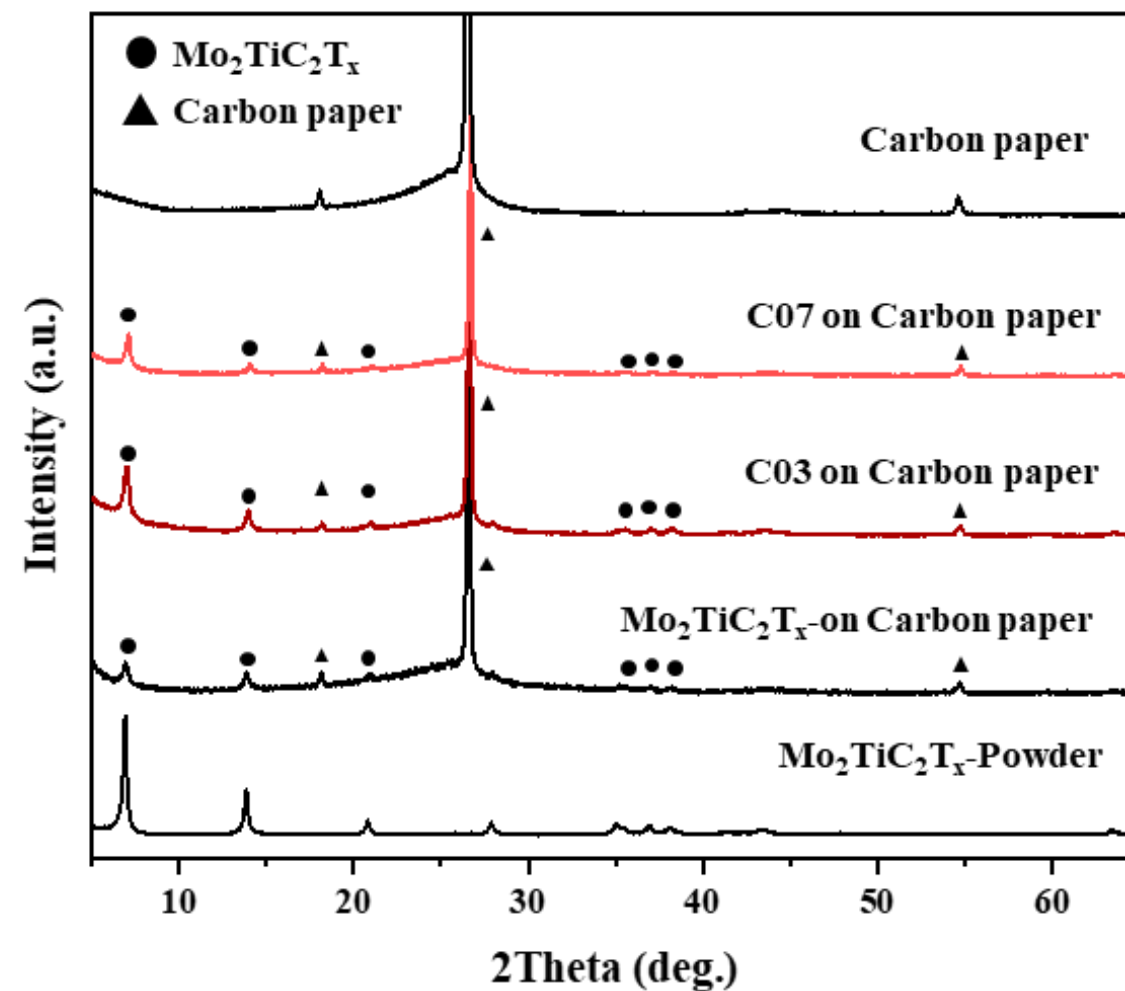


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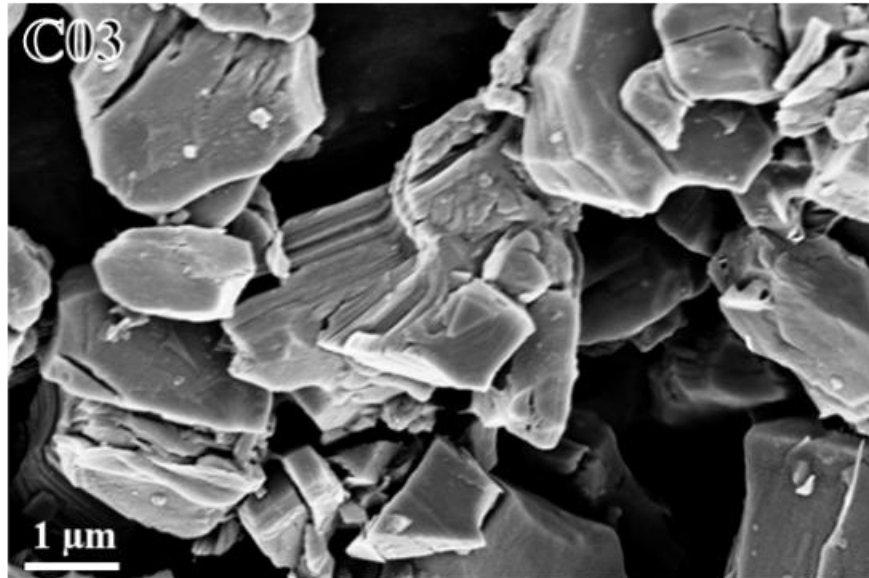




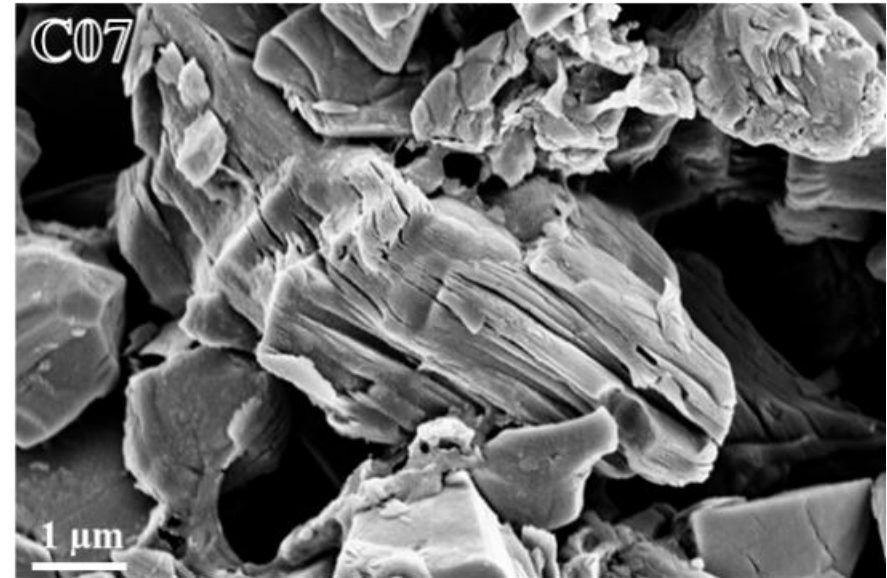


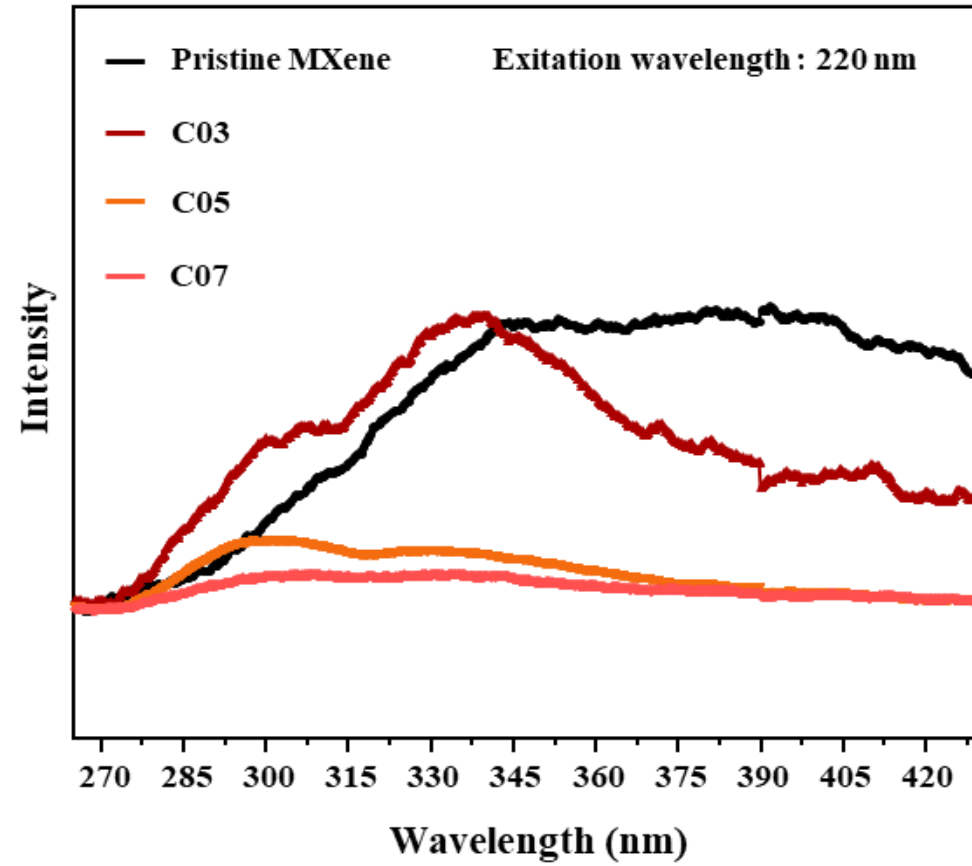


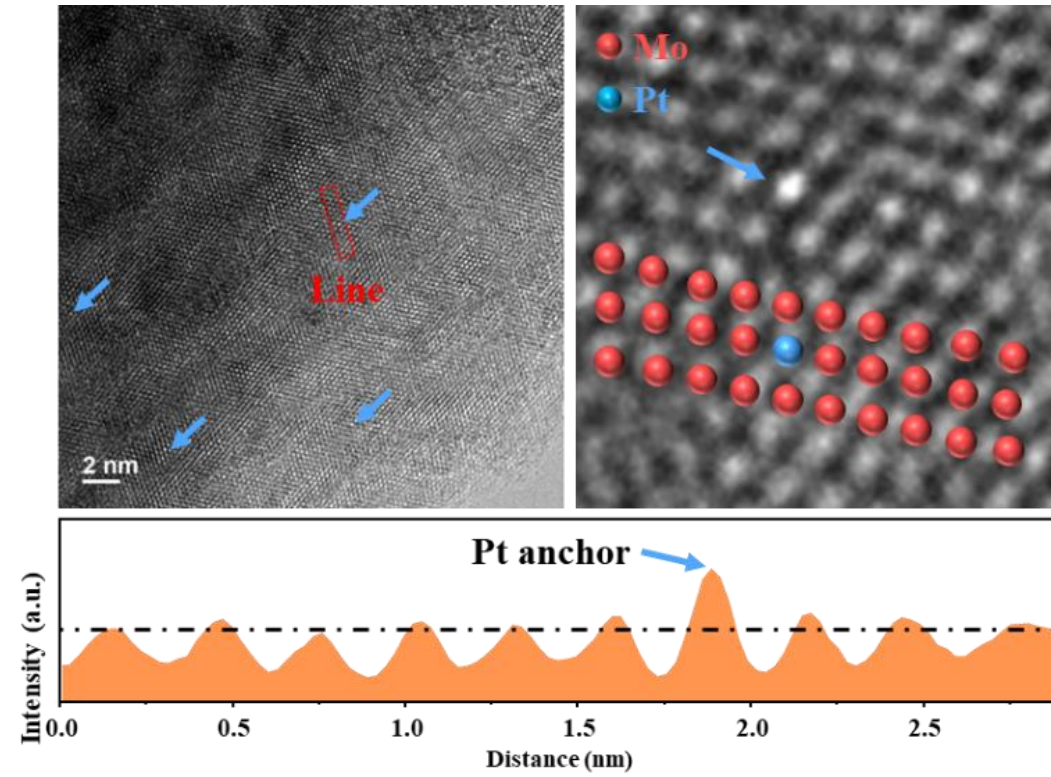
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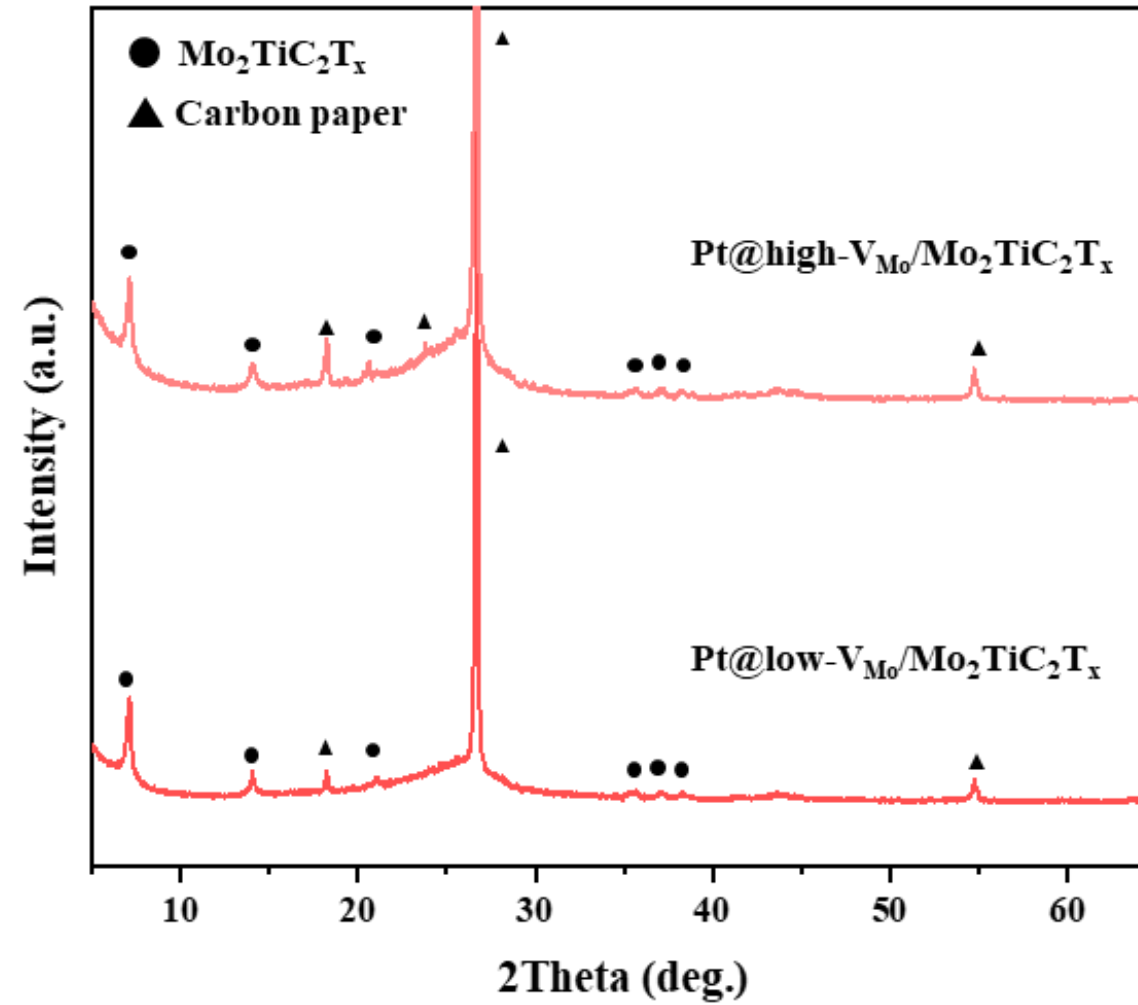


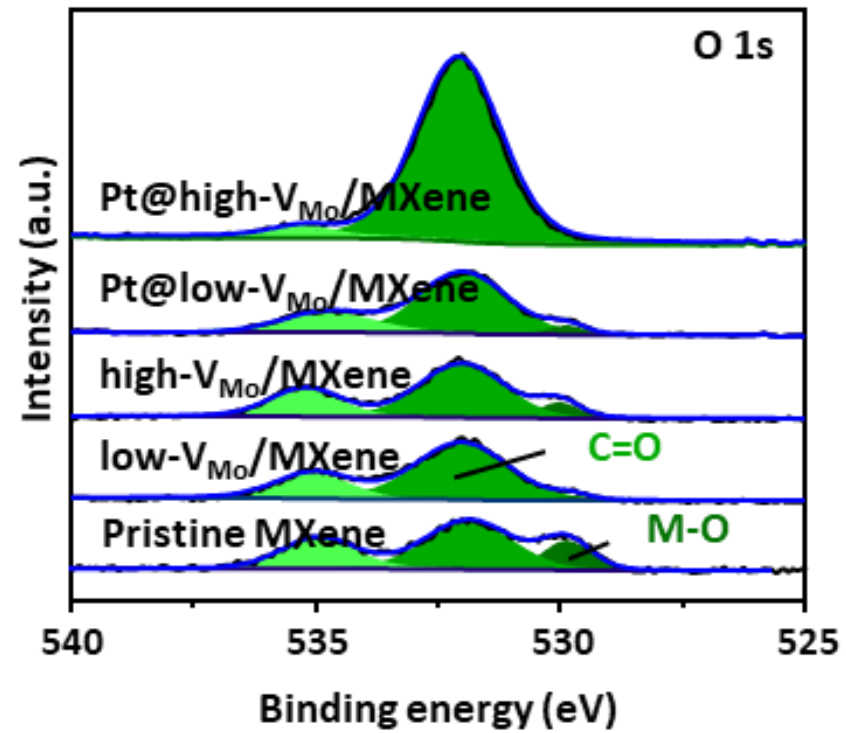
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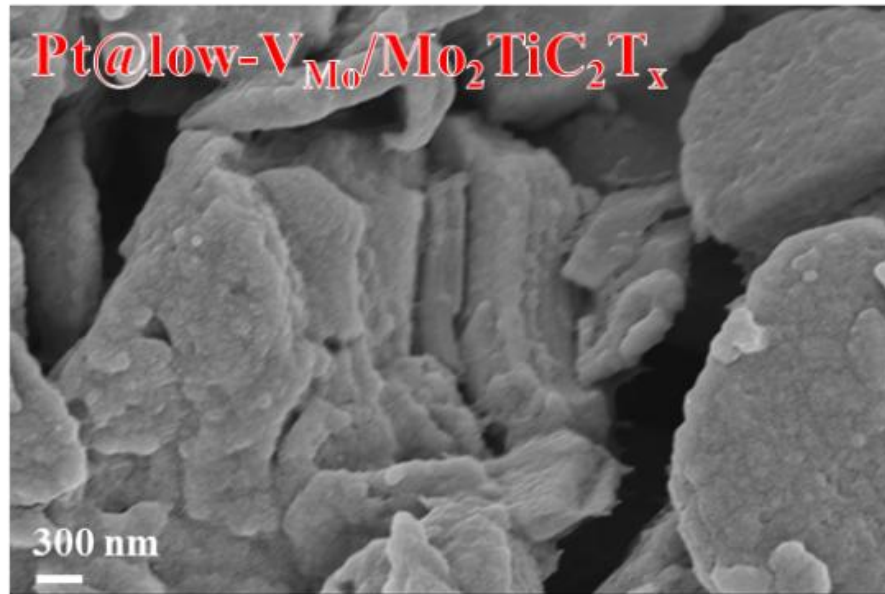




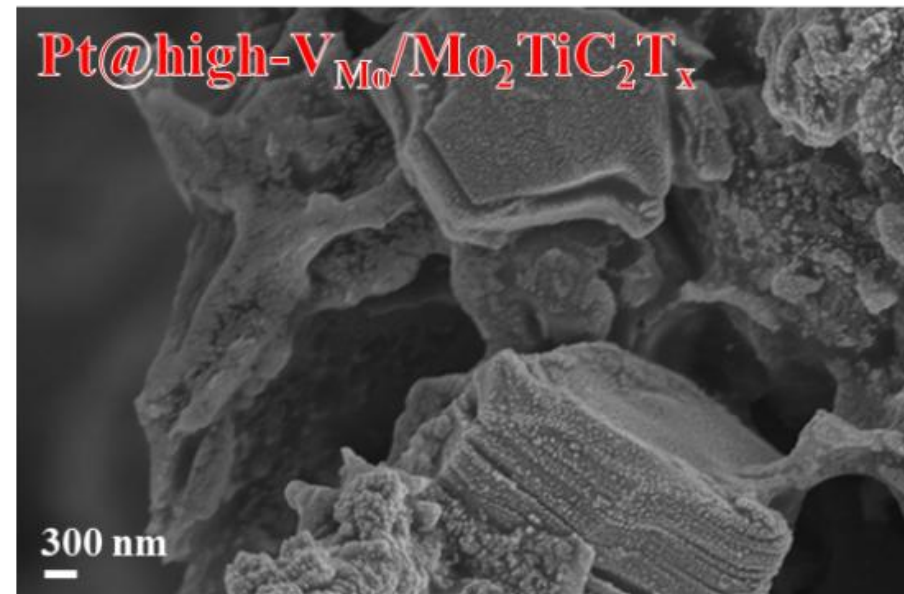




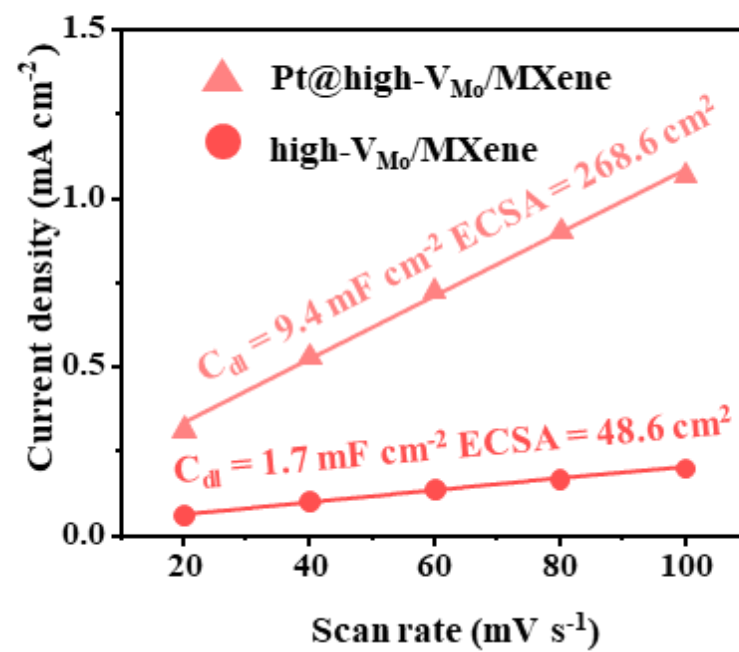
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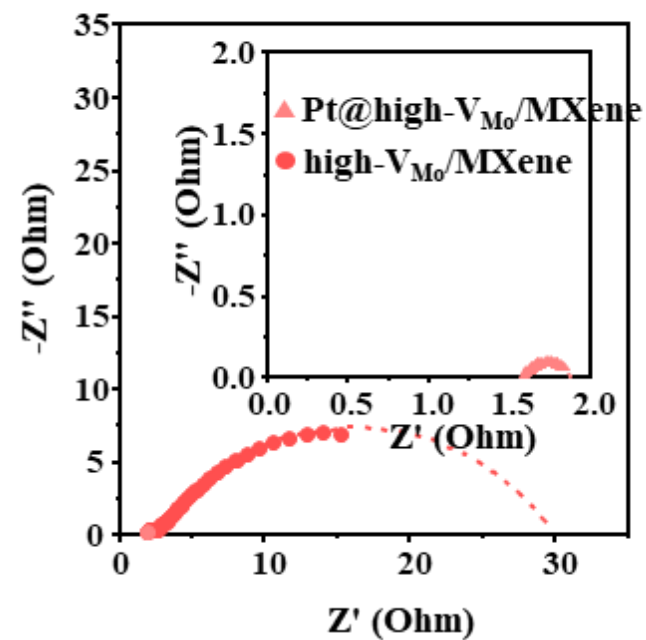
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a



b



Stability test:

